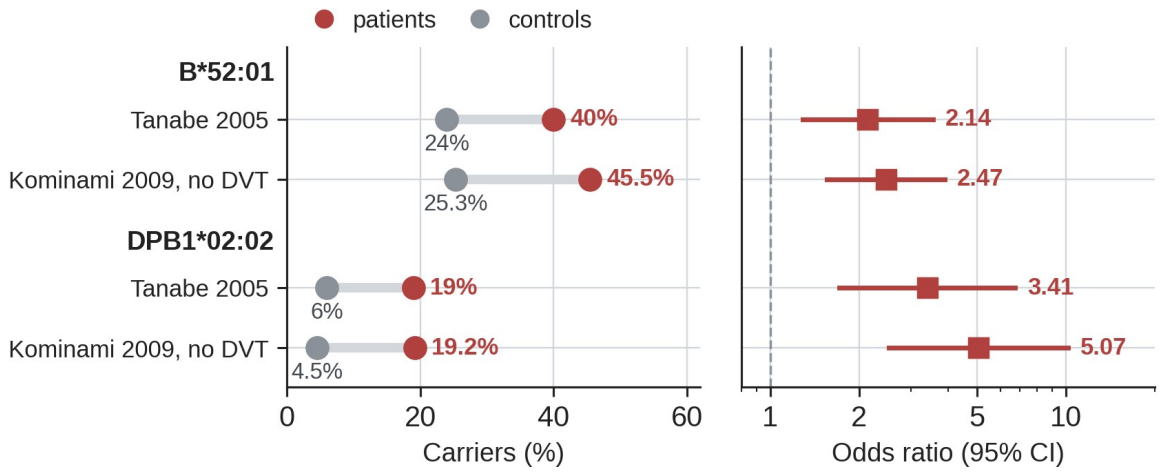
Chronic Thromboembolic Pulmonary Hypertension (CTEPH):
HLA Typing from Whole Genomes in JHRP v6

ZHAO TIE
2026-10-19

HLA Associations with Disease, Including CTEPH

- The HLA region is associated with more diseases, mainly autoimmune and infectious, than any other region of the genome [1].
- Classic examples: ankylosing spondylitis with HLA-B27 and celiac disease with HLA-DQB1*02 [2]; severe skin reactions to carbamazepine with HLA-B*15:02 in Han Chinese [3].

HLA alleles reported in Japanese CTEPH



As reported, none recomputed: Tanabe [4], all CTEPH, 80 patients / 678 controls; Kominami [5], CTEPH without deep vein thrombosis, 99 / 380, 59 of them among Tanabe's 80.

This study: full mainland cohort

Group	Whole genomes
CTEPH cases	439
AGP3K controls	2,662
Total	**3,101**

Testing such alleles requires the HLA alleles of every participant. In a sequenced cohort these are inferred from the reads, so they must be checked before use.

HLA alleles are associated with many diseases, including CTEPH in Japanese patients.

Abbreviations: CI, confidence interval; CTEPH, chronic thromboembolic pulmonary hypertension; HLA, human leukocyte antigen.

[1] Trowsdale, J., & Knight, J. C. (2013). Major histocompatibility complex genomics and human disease. *Annual Review of Genomics and Human Genetics, 14*, 301–323.
[2] Choo, S. Y. (2007). The HLA system: Genetics, immunology, clinical testing, and clinical implications. *Yonsei Medical Journal, 48*(1), 11–23.
[3] Chung, W. H., Hung, S. I., Hong, H. S., Hsih, M. S., Yang, L. C., Ho, H. C., ... & Chen, Y. T. (2004). Medical genetics: A marker for Stevens-Johnson syndrome. *Nature, 428*(6982), 486.
[4] Tanabe, N., Kimura, A., Amano, S., Okada, O., Kasahara, Y., Tatsumi, K., ... & Kuriyama, T. (2005). Association of clinical features with HLA in chronic pulmonary thromboembolism. *European Respiratory Journal, 25*(1), 131–138.
[5] Kominami, S., Tanabe, N., Ota, M., Naruse, T. K., Katsuyama, Y., Nakanishi, N., ... & Kimura, A. (2009). HLA-DPB1 and NFKBIL1 may confer the susceptibility to chronic thromboembolic pulmonary hypertension in the absence of deep vein thrombosis. *Journal of Human Genetics, 54*(2), 108–114.

Aims and Outline

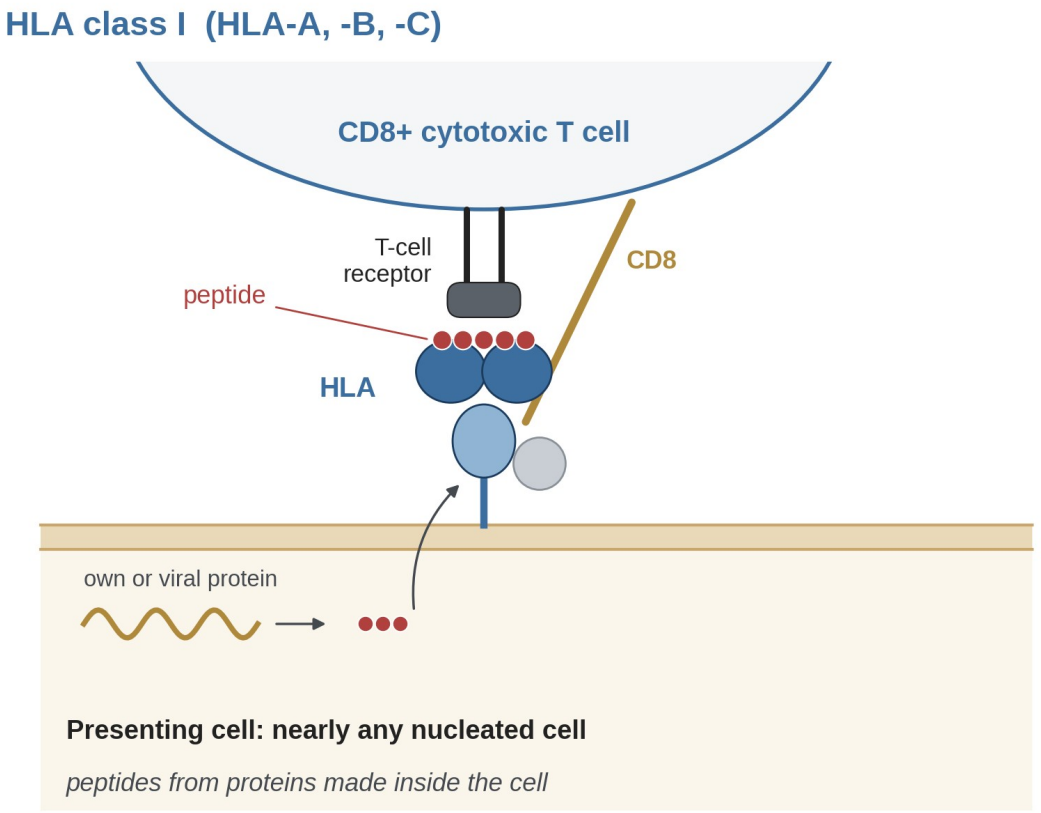
Question. Are the HLA genotypes typed from the whole genomes of this cohort complete and consistent with the Japanese population, and which HLA markers can a test of association with CTEPH then use?

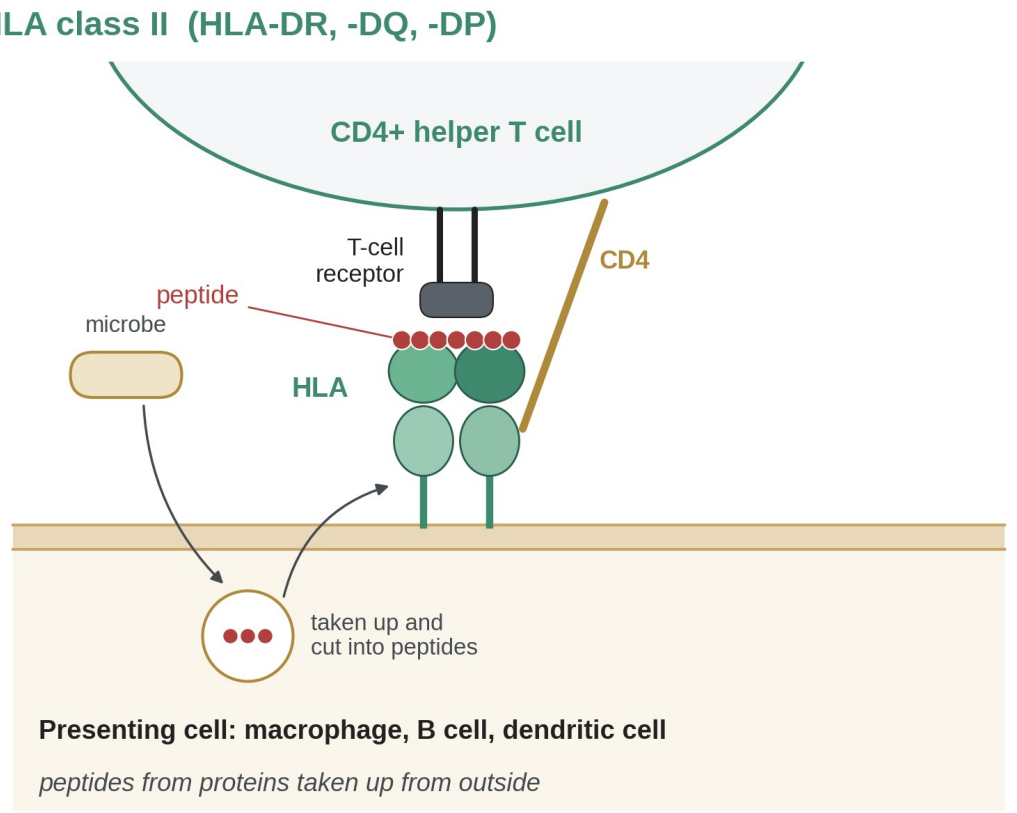
Part	Aim	Evidence presented
Background	Introduce the HLA system, its nomenclature, and why typing it from reads is difficult	structure, diversity, allele names, DR haplotypes
Method	Infer 2-field genotypes of 19 HLA genes from the whole-genome reads, and make them comparable with the Japanese reference	HLA-HD; *DRB3*, *DRB4*, and *DRB5* checked against *DRB1*; the two criteria; harmonization with ToMMo 61KJPN
Quality	Show that the genotypes are complete and consistent with the Japanese population	completeness; presence of the called alleles in ToMMo; frequency concordance
Markers	Encode alleles and amino-acid residues as markers and filter them, in three cohorts	call rate ≥ 95% in cases and in controls, then MAF ≥ 0.5%; stepwise record
Summary	State what the association analysis can use	methods statement; caveat (detail in the Notes)

Cohort: 3,101 whole genomes of the full mainland cohort, 439 CTEPH cases and 2,662 AGP3K controls; two nested subsets (narrow, intermediate) are carried through the marker filters.

Before any association test, the typed genotypes must be shown complete and population-consistent.

Function of the HLA Molecules





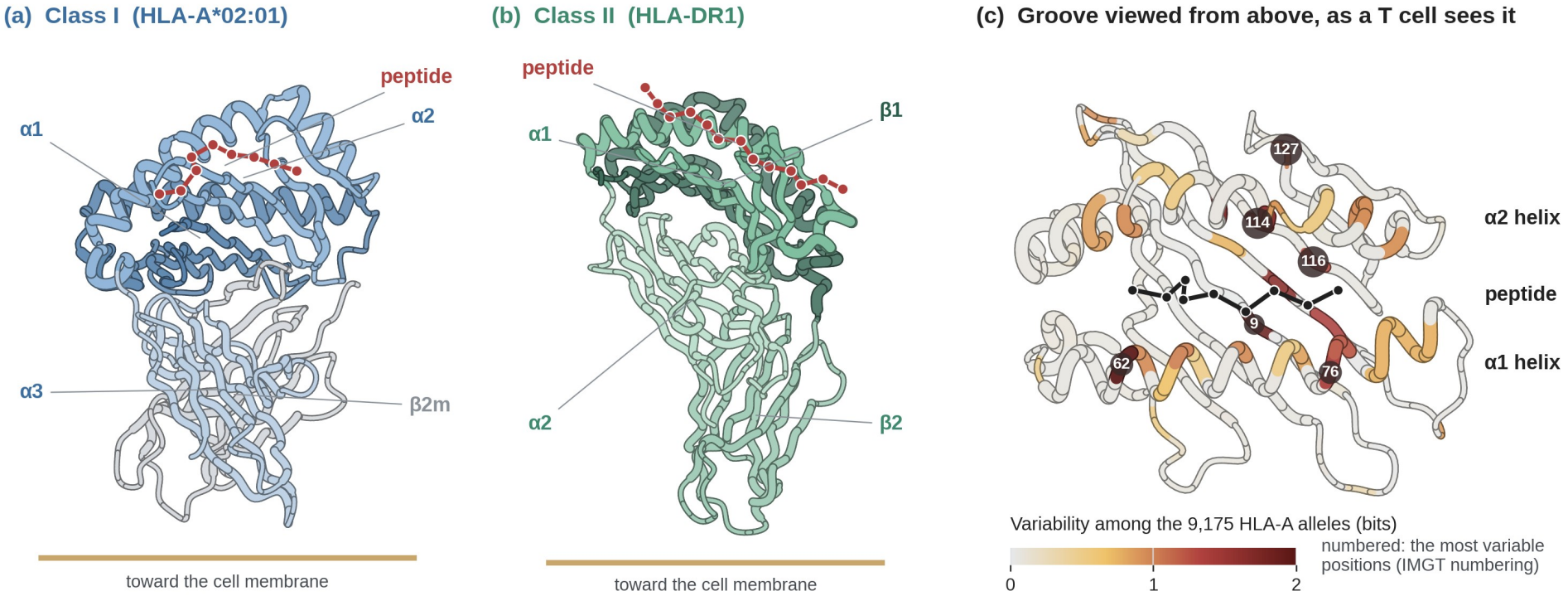
- HLA (human leukocyte antigen) molecules carry short protein fragments, peptides, to the cell surface, where T cells inspect them [1].
- Class I (*HLA-A*, *-B*, *-C*) is expressed on nearly all nucleated cells and presents peptides from inside the cell to CD8+ T cells; class II (*HLA-DR*, *-DQ*, *-DP*) is expressed on antigen-presenting cells and presents peptides from outside to CD4+ T cells [1].
- Allelic variation changes the amino acids that bind the peptide, and with them the peptides a molecule can present [1].

Which peptides reach a person's T cells depends on the HLA alleles that person carries.

Abbreviations: CD4, T-cell marker of helper T cells; CD8, T-cell marker of cytotoxic T cells; HLA, human leukocyte antigen.

[1] Medhasi, S., & Chantratita, N. (2022). Human leukocyte antigen (HLA) system: Genetics and association with bacterial and viral infections. *Journal of Immunology Research, 2022*, 9710376.

Structure of the HLA Molecules and the Site of Allelic Variation



Structures: PDB 1HHK, HLA-A*02:01 with a viral peptide [4]; PDB 1DLH, HLA-DR1 with an influenza peptide [5]. Variability: Shannon entropy of the amino acids at each position across the 9,175 HLA-A alleles of IPD-IMGT/HLA 3.64.0 [6] (0 = identical in all alleles).

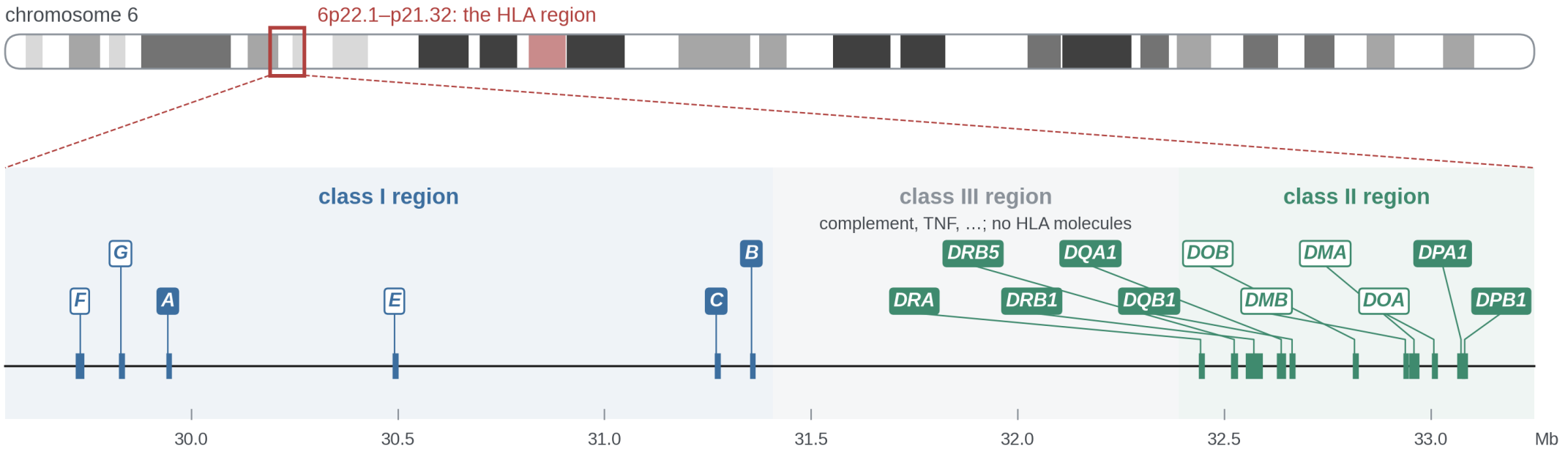
- Class I: a heavy chain (domains α1, α2, α3) with β2-microglobulin; α1 and α2 form the peptide-binding groove. Class II: an α and a β chain; α1 and β1 form the groove [1].
- Most HLA variation lies in the exons that encode the groove [2]: exons 2 and 3 in class I, exon 2 in class II [3]. In HLA-A, the most variable positions line the floor and the helices of the groove (panel c).

Allelic variation is concentrated in the groove, which determines the peptides a molecule can bind.

Abbreviations: HLA, human leukocyte antigen; IPD-IMGT/HLA, Immuno Polymorphism Database – international ImMunoGeneTics information system, HLA section; PDB, Protein Data Bank.

[1] Medhasi, S., & Chantratita, N. (2022). Human leukocyte antigen (HLA) system: Genetics and association with bacterial and viral infections. *Journal of Immunology Research, 2022*, 9710376.
[2] Trowsdale, J., & Knight, J. C. (2013). Major histocompatibility complex genomics and human disease. *Annual Review of Genomics and Human Genetics, 14*, 301–323.
[3] Anthony Nolan Research Institute. (n.d.). *P groups*. HLA Nomenclature. Retrieved October 19, 2026, from https://hla.alleles.org/pages/p&g_groups/p_groups/
[4] Madden, D. R., Garboczi, D. N., & Wiley, D. C. (1993). The antigenic identity of peptide-MHC complexes: A comparison of the conformations of five viral peptides presented by HLA-A2. *Cell, 75*(4), 693–708.
[5] Stern, L. J., Brown, J. H., Jardetzky, T. S., Gorga, J. C., Urban, R. G., Strominger, J. L., & Wiley, D. C. (1994). Crystal structure of the human class II MHC protein HLA-DR1 complexed with an influenza virus peptide. *Nature, 368*(6468), 215–221.
[6] Barker, D. J., Maccari, G., Georgiou, X., Cooper, M. A., Flicek, P., Robinson, J., & Marsh, S. G. E. (2023). The IPD-IMGT/HLA Database. *Nucleic Acids Research, 51*(D1), D1053–D1060.

The HLA Region on Chromosome 6



Filled labels: genes of the classical molecules HLA-A, -B, -C, -DR, -DQ, and -DP, which present peptides to T cells. Open labels: other HLA genes.

DRB3 and DRB4 are not on the GRCh38 primary chromosome 6, which carries a DR haplotype with DRB5. Red box: reads extracted for typing (29.54–33.42 Mb).

Positions: Ensembl, GRCh38; bands: UCSC hg38.

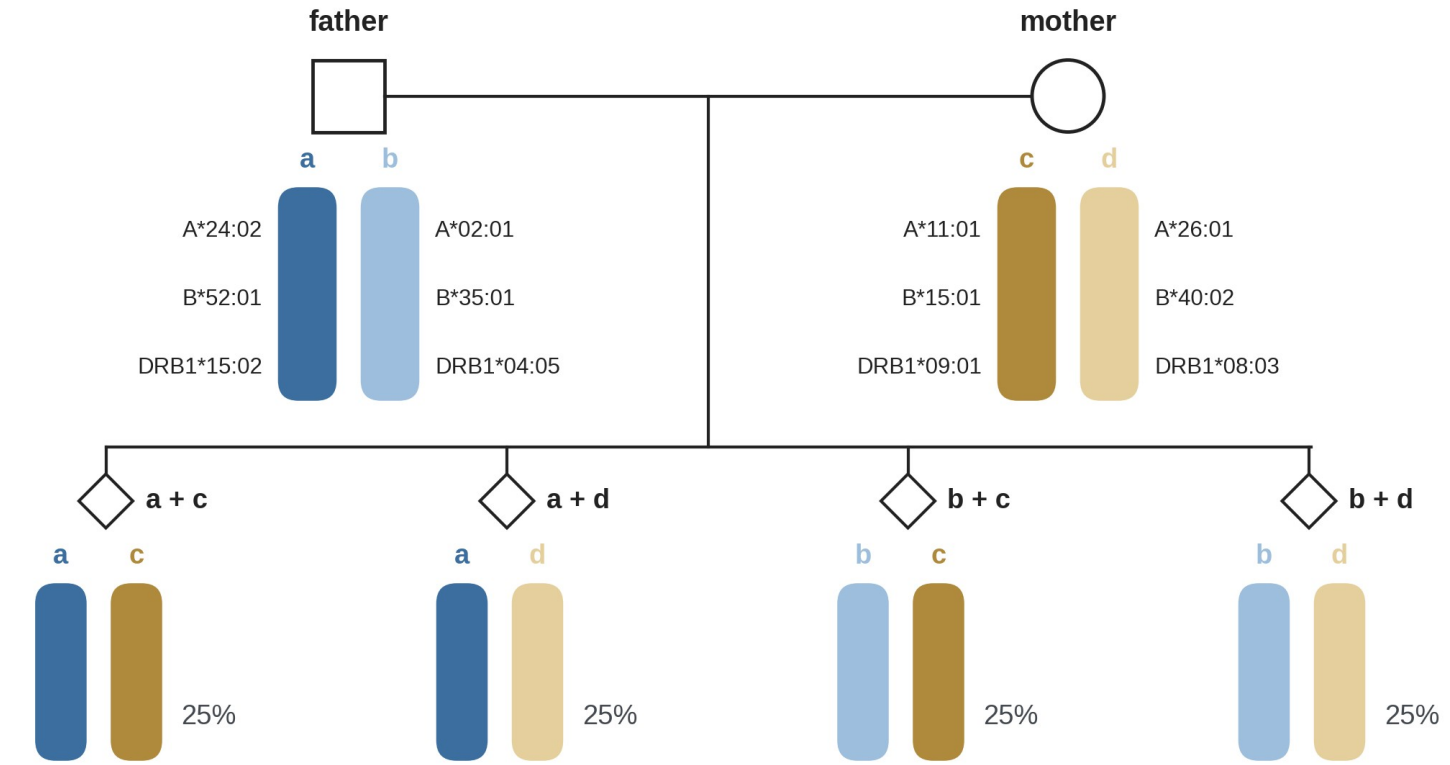
- The HLA region, or major histocompatibility complex, holds about 260 genes in about 4 Mb [1]. Class I genes lie at one end and class II genes at the other; the class III region between them holds genes such as complement and TNF, but no HLA molecules [2], [3].
- Class II molecules pair two chains: *DQA1* + *DQB1* form DQ, *DPA1* + *DPB1* form DP, and the invariant DR α chain (*DRA*) pairs with the β chain of *DRB1*, *DRB3*, *DRB4*, or *DRB5* [3].

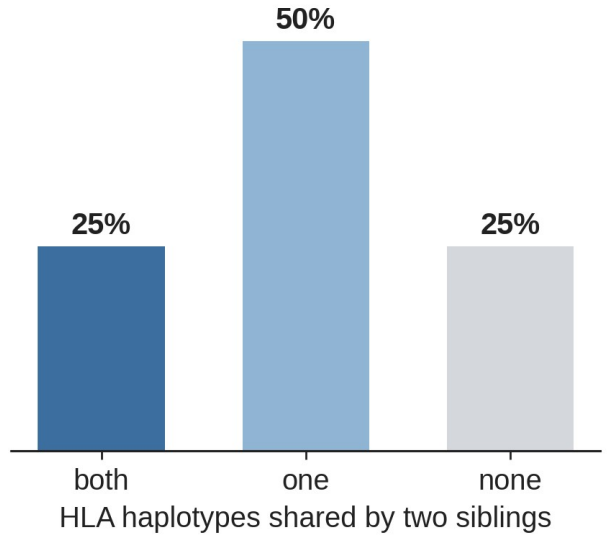
The HLA genes lie together in one 4-Mb region of chromosome 6: class I at one end, class II at the other.

Abbreviations: HLA, human leukocyte antigen; Mb, million bases; TNF, tumor necrosis factor.

[1] Trowsdale, J., & Knight, J. C. (2013). Major histocompatibility complex genomics and human disease. *Annual Review of Genomics and Human Genetics, 14*, 301–323.
[2] Medhasi, S., & Chantratita, N. (2022). Human leukocyte antigen (HLA) system: Genetics and association with bacterial and viral infections. *Journal of Immunology Research, 2022*, 9710376.
[3] Choo, S. Y. (2007). The HLA system: Genetics, immunology, clinical testing, and clinical implications. *Yonsei Medical Journal, 48*(1), 11–23.

Inheritance of the HLA Haplotypes





Alleles are examples. Each bar is one chromosome 6; its color marks the haplotype it carries.

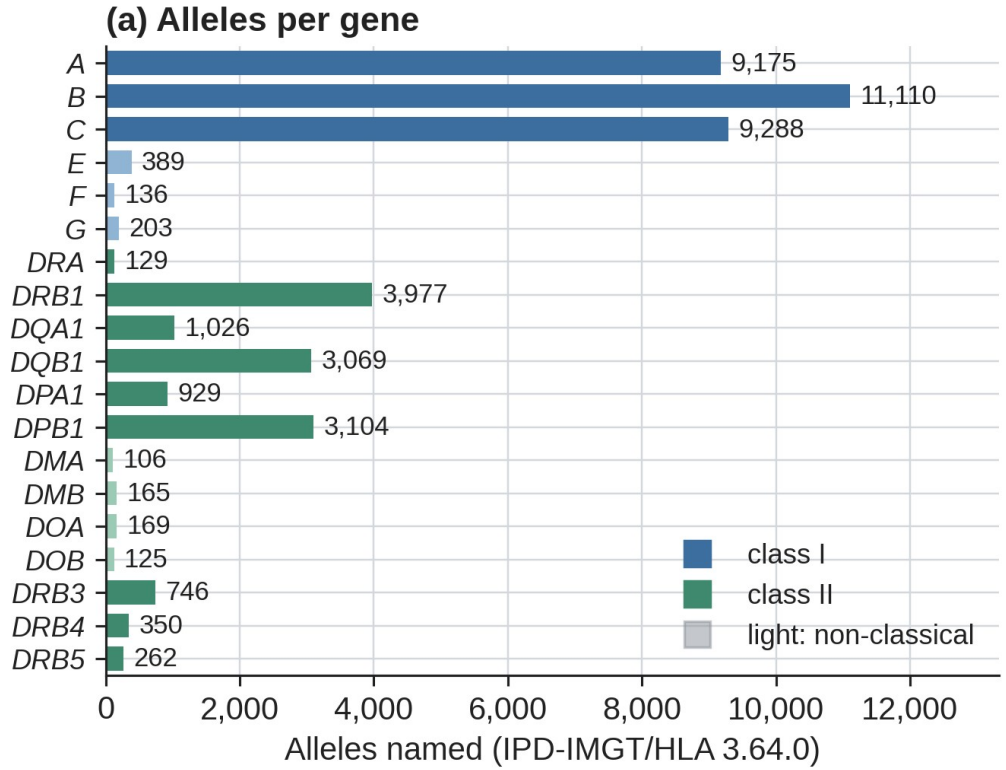
- HLA genes are closely linked, so the region passes from each parent as one block, a haplotype; two siblings share both haplotypes with a probability of 25%, one with 50% and none with 25% [1].
- A person's HLA genotype is therefore a pair of alleles at each gene, e.g., A*24:02 / A*11:01, which may be identical (homozygous). Some haplotypes are more common than chance predicts (linkage disequilibrium) [1].

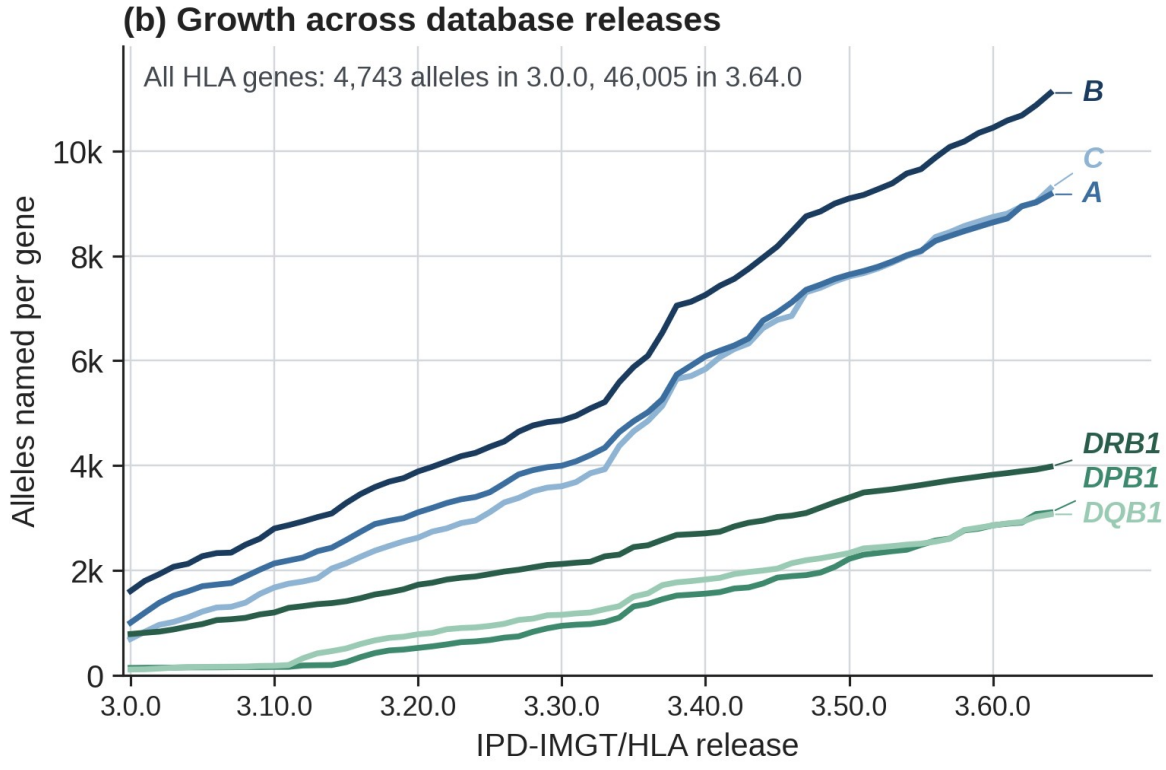
Each person carries two HLA haplotypes, one from each parent, and so two alleles of every HLA gene.

Abbreviations: HLA, human leukocyte antigen.

[1] Choo, S. Y. (2007). The HLA system: Genetics, immunology, clinical testing, and clinical implications. *Yonsei Medical Journal, 48*(1), 11–23.

Allelic Diversity of the HLA Genes





- Most human genes have two or three common alleles; *HLA-B*, the most polymorphic, already had more than 2,000 by 2012 [1]. Release 3.64.0 of IPD-IMGT/HLA names 46,005 HLA alleles, 11,110 of them for *HLA-B* (panel a).
- The number of named alleles has grown about 10-fold since release 3.0.0 (4,743 → 46,005); next-generation sequencing has driven this increase [2] (panel b).

No other human genes carry as many alleles, which is what makes HLA typing difficult.

Abbreviations: HLA, human leukocyte antigen; IPD-IMGT/HLA, Immuno Polymorphism Database – international ImMunoGeneTics information system, HLA section.

[1] Trowsdale, J., & Knight, J. C. (2013). Major histocompatibility complex genomics and human disease. *Annual Review of Genomics and Human Genetics, 14*, 301–323.
[2] Barker, D. J., Maccari, G., Georgiou, X., Cooper, M. A., Flicek, P., Robinson, J., & Marsh, S. G. E. (2023). The IPD-IMGT/HLA Database. *Nucleic Acids Research, 51*(D1), D1053–D1060.

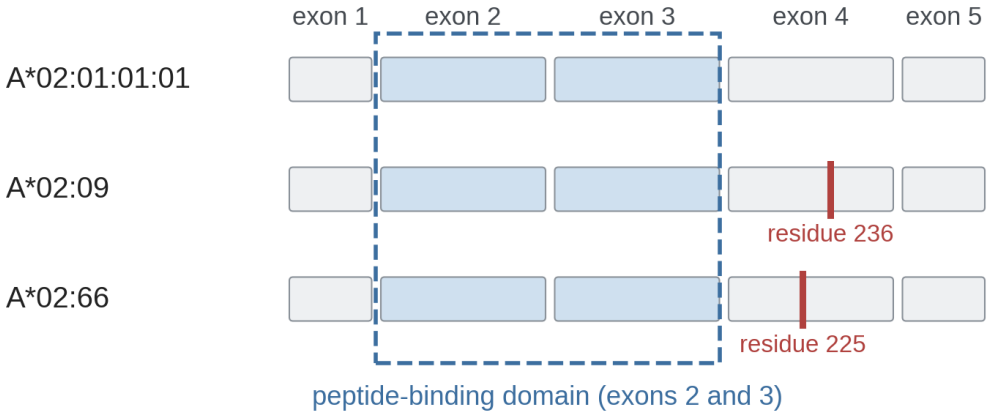
Allele Nomenclature and Typing Resolution

2-field resolution = the protein



gene	HLA-A, HLA-B, …
1	allele group (often the serological antigen)
2	a distinct protein (amino-acid sequence)
3	synonymous change in the coding sequence
4	change outside the coding sequence
N	suffix: not expressed (null allele)

P group: identical protein where the peptide is held



The three proteins differ only outside exons 2 and 3, so they form one P group, A*02:01P (IPD-IMGT/HLA 3.64.0). ToMMo reports alleles at this resolution.

- Fields are separated by colons. Alleles that differ in the first two fields differ in amino-acid sequence; a third field marks a synonymous change, a fourth a change outside the coding sequence, and N a null allele [1].
- A P group joins alleles that encode the same protein over exons 2 and 3 (class I) or exon 2 (class II), the peptide-binding domain, and is named after its lowest-numbered member with a P (A*02:01P) [2]. An allele whose protein there is shared by no other allele, and every null allele, belongs to no P group and keeps its own name; at P-group resolution each allele is replaced by its P group, if it has one [2].

Genotypes are reported at 2-field (protein) resolution and compared with ToMMo at P-group resolution.

Abbreviations: HLA, human leukocyte antigen; ToMMo, Tohoku Medical Megabank Organization.

[1] Marsh, S. G., Albert, E. D., Bodmer, W. F., Bontrop, R. E., Dupont, B., Erlich, H. A., ... & Trowsdale, J. (2010). Nomenclature for factors of the HLA system, 2010. *Tissue Antigens, 75*(4), 291–455.
[2] Anthony Nolan Research Institute. (n.d.). *P groups*. HLA Nomenclature. Retrieved October 19, 2026, from https://hla.alleles.org/pages/p&g_groups/p_groups/

DR Haplotypes and the DRB Genes

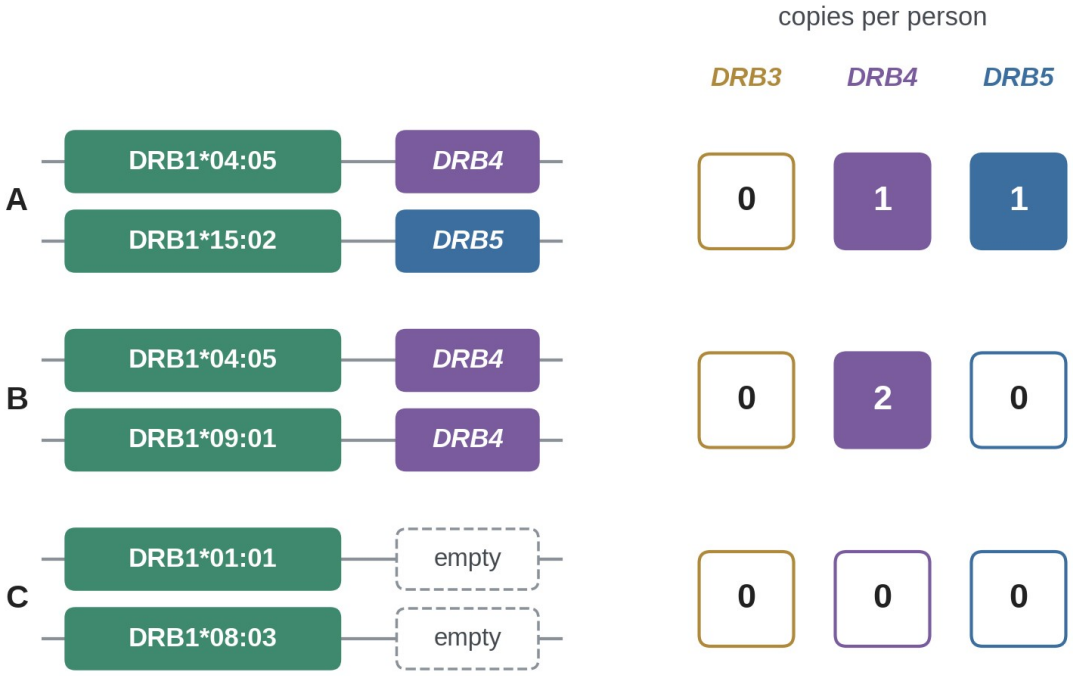
One chromosome 6

slot 1 always holds DRB1; its group decides slot 2

DRB1 group	slot 1	slot 2	haplotype
DRB1*15, *16	*DRB1*	*DRB5*	**DR51**
DRB1*03, *11, *12, *13, *14	*DRB1*	*DRB3*	**DR52**
DRB1*04, *07, *09	*DRB1*	*DRB4*	**DR53**
DRB1*01, *08, *10	*DRB1*	empty	**DR1, DR8, DR10**

DRA, on every chromosome, is not drawn. Exceptions to the rule occur.

One person: two chromosomes, add up



Person C carries none of the three: 0 copies is a correct result, not a failed typing.

- Every chromosome 6 carries *DRB1*; whether it also carries *DRB3*, *DRB4*, or *DRB5* depends on the group of that *DRB1* allele [1], [2] (left), with occasional exceptions [2].
- A person has two chromosomes, so *DRB3*, *DRB4*, and *DRB5* are each present in 0, 1, or 2 copies (right); 0 copies is a correct result, not a failed typing.

The two *DRB1* alleles determine how many copies of *DRB3*, *DRB4*, and *DRB5* a person carries.

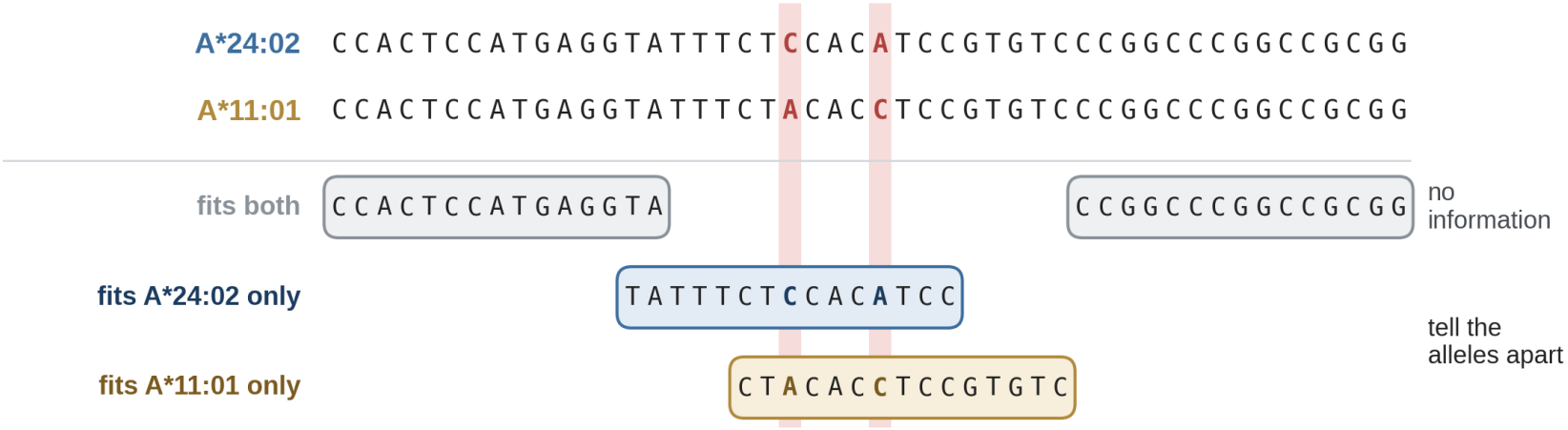
Abbreviations: HLA, human leukocyte antigen.

[1] Andersson, G. (1998). Evolution of the human HLA-DR region. *Frontiers in Bioscience, 3*, d739–d745.
[2] Anthony Nolan Research Institute. (n.d.). *DRB haplotypes*. HLA Nomenclature. Retrieved October 19, 2026, from https://hla.alleles.org/pages/other/drb_haplotypes/

Typing from Sequencing Reads

One person's two HLA-A alleles

48 bases of exon 2, real sequence; the alleles differ at 2 of them (red)



Each box is a read, aligned under the alleles; reads are shortened for display (real reads are about 150 bases).

A look-alike gene

schematic



A read from the pseudogene HLA-Y fits HLA-A as well: it can be counted for the wrong gene and create a false allele.

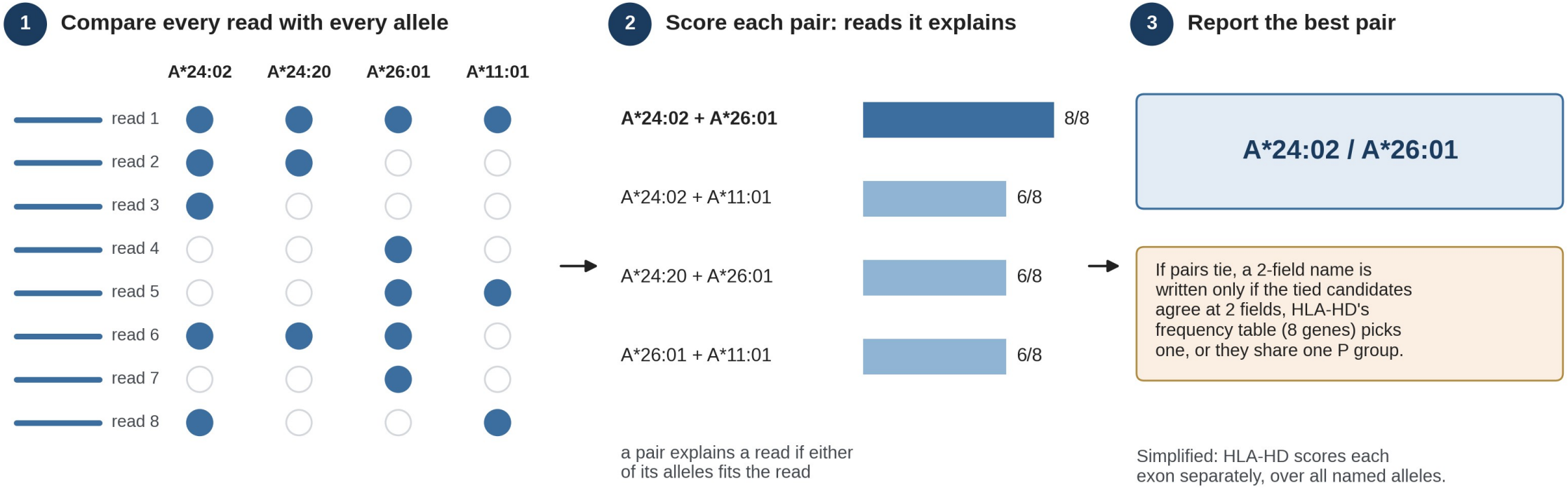
- A read is a short stretch of sequenced DNA, about 150 bases. The typing reuses the whole-genome reads of the HLA region, about 100–170 per gene and person (median over heterozygous calls); no laboratory assay is needed.
- Two alleles of a gene differ at few positions, so many of a person's reads fit both and carry no information (median 22–52% at the classical genes, 80–95% at the others); only reads over a difference tell them apart (left). Reads of a look-alike gene, such as the pseudogene *HLA-Y*, can be counted for the wrong gene [1] (right).

Only reads spanning a difference distinguish alleles, so all reads are weighed against all candidate alleles at once.

Abbreviations: HLA, human leukocyte antigen.

[1] Kawaguchi, S., Higasa, K., Shimizu, M., Yamada, R., & Matsuda, F. (2017). HLA-HD: An accurate HLA typing algorithm for next-generation sequencing data. *Human Mutation, 38*(7), 788–797.

The HLA-HD Typing Algorithm



- HLA-HD compares every read with every named allele in the IPD-IMGT/HLA dictionary, scores each allele pair by the reads it explains, exon by exon, and reports the best pair [1]. Run once over the 3,101 genomes (HLA-HD 1.7.1, IPD-IMGT/HLA 3.64.0); 19 genes reported.
- A 2-field name is written only when it is settled: the tied candidates agree at 2 fields (93.4% of the 107,944 alleles called), HLA-HD's frequency table picks one (5.3%), or they share one P group (0.5%). Otherwise none is written (0.8%); no allele is picked arbitrarily.

HLA-HD reports the allele pair that best explains the reads; a 2-field name is written only when settled.

Abbreviations: HLA, human leukocyte antigen; IPD-IMGT/HLA, Immuno Polymorphism Database – international ImMunoGeneTics information system, HLA section.

[1] Kawaguchi, S., Higasa, K., Shimizu, M., Yamada, R., & Matsuda, F. (2017). HLA-HD: An accurate HLA typing algorithm for next-generation sequencing data. *Human Mutation, 38*(7), 788–797.

Validation of DRB3, DRB4, and DRB5 Copy Number against DRB1

Rationale. HLA-HD types *DRB3*, *DRB4*, and *DRB5* from the reads alone, without looking at *DRB1*. The two *DRB1* alleles predict how many copies of each a person carries. For every person the two answers are compared; one real case of each outcome is shown.

Outcome	Example: *DRB1*	*DRB1* predicts	HLA-HD found	Recorded	*DRB3*	*DRB4*	*DRB5*
Agree	DRB1*15:02 / DRB1*04:05	1 *DRB4*, 1 *DRB5*	DRB4*01:03 and DRB5*01:02	as found, one copy each	2,599 (84%)	2,725 (88%)	3,007 (97%)
One allele too many	DRB1*15:02 / DRB1*09:01	1 *DRB5*	DRB5*01:02 (55 reads of its own) and DRB5*01:08 (1)	DRB5*01:02, one copy: the allele with more reads of its own	367	327	86
A gene that should be absent	DRB1*09:01 / DRB1*08:03	no *DRB3*	DRB3*02:131 (5 reads)	no *DRB3* (0 copies)	128	14	4
A gene not found	DRB1*04:05 / DRB1*08:02	1 *DRB4*	nothing	no genotype (left blank)	5	33	2

Last three columns: number of people with each outcome, of the 3,099 with a *DRB1* genotype (2 of the 3,101 have none).

Why *DRB1* is decisive. An allele is supported by reads of its own: reads that fit it and not the other allele. An extra allele reported by HLA-HD usually has almost none (1 against 55 in the example): it holds under 15% of these reads in 453 of 660 extra-allele calls (69%), against 120 of 396 (30%) where two copies are expected.

In the exported matrix: 0, 1, or 2 copies, matching the *DRB1* prediction in every person who has one (*DRB3* 3,049, *DRB4* 2,835, *DRB5* 3,071); the 2 per gene without a *DRB1* genotype keep HLA-HD's call. *DRB3*, *DRB4*, and *DRB5* are not used in the association analysis.

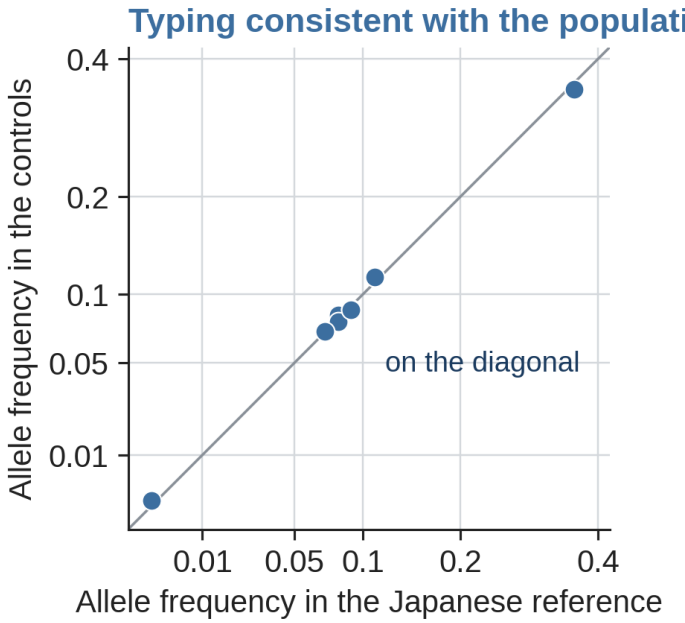
HLA-HD agrees with *DRB1* in 84–97% of people; where it does not, the *DRB1* rule decides.

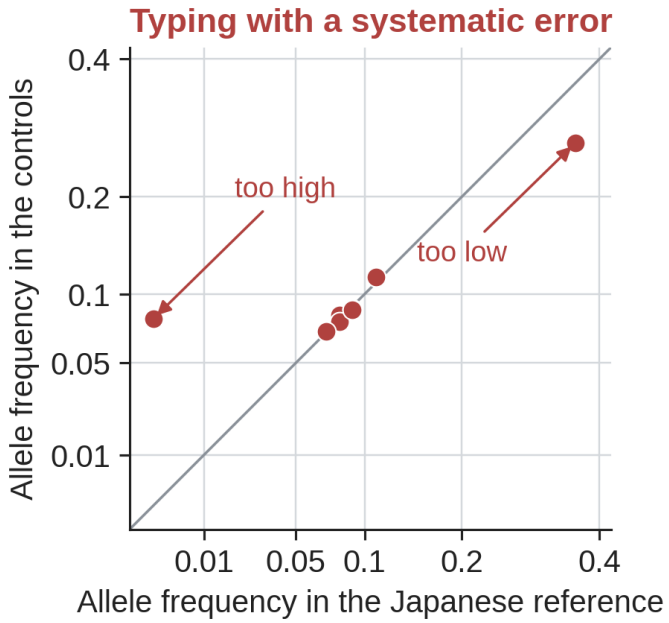
Abbreviations: HLA, human leukocyte antigen.

[1] Anthony Nolan Research Institute. (n.d.). *DRB haplotypes*. HLA Nomenclature. Retrieved October 19, 2026, from https://hla.alleles.org/pages/other/drb_haplotypes/

Criteria for Assessing Typing Quality

illustration





Check 1: completeness	Check 2: population agreement
the share of people with a 2-field genotype, gene by gene	the alleles of the controls are found in 61,424 Japanese individuals (ToMMo 61KJPN) [1], at the same frequencies

r: Pearson correlation of the allele frequencies of one gene in the two sources; 1 means the same frequency pattern. Only controls are compared, because the cases are selected for disease.

A systematic typing error would shift allele frequencies away from the Japanese reference.

Abbreviations: 61KJPN, about 61,000 Japanese individuals; ToMMo, Tohoku Medical Megabank Organization.

[1] Tadaka, S., Kawashima, J., Hishinuma, E., Saito, S., Okamura, Y., Otsuki, A., ... & Kinoshita, K. (2024). jMorp: Japanese Multi-Omics Reference Panel update report 2023. *Nucleic Acids Research, 52*(D1), D622–D632.

Making Our Calls and the ToMMo Table Comparable

Step	Our controls HLA-HD calls, IPD-IMGT/HLA 3.64.0	ToMMo 61KJPN the published table [1]	Bias prevented if the step were skipped
1 Comparable population	2,662 AGP3K controls of the full mainland cohort (mainland Japanese ancestry); the 439 cases excluded	61,424 individuals of the Tohoku Medical Megabank cohorts (Miyagi and Iwate, mainland Japan)	differences due to disease or ancestry taken for typing error
2 Same genes	the 11 genes reported by both: *A*, *B*, *C*, *E*, *F*, *G*, *DRB1*, *DQA1*, *DQB1*, *DPA1*, *DPB1*	11 of its 13 genes; *DRB3* and *DRB4* excluded, as its entries for them are inconsistent with its own *DRB1*	*DRB3* and *DRB4* differing for reasons unrelated to typing
3 Same resolution	each 2-field allele replaced by its P group, where one exists [2]: 1,682 alleles become 1,295	published at this resolution: 440 P groups and 913 alleles in no P group (e.g., B*39:04); read with the same P-group file	alleles without a counterpart: DQA1*03:03 (792 copies) is listed by ToMMo only within DQA1*03:01P
4 Same frequency definition	allele frequency = copies of the allele / protein-coding copies at the gene; null alleles excluded; a control counted only when both copies are typed to 2 fields	null alleles excluded (e.g., A*24:86N); frequencies recomputed over the protein-coding copies, so they add up to 100%	ToMMo frequencies deflated by its null entries; under-representation of alleles that are harder to type
5 Dictionary release	497 of the 1,295 alleles were named after release 3.21.0	all its alleles were named by release 3.21.0	those 497 counted as absent from ToMMo, although it could not contain them; reported separately instead

Result, per gene. One list of alleles at P-group resolution, each with its frequency among the protein-coding copies on each side. The Quality part compares these lists: which of our alleles ToMMo carries, and at what frequencies.

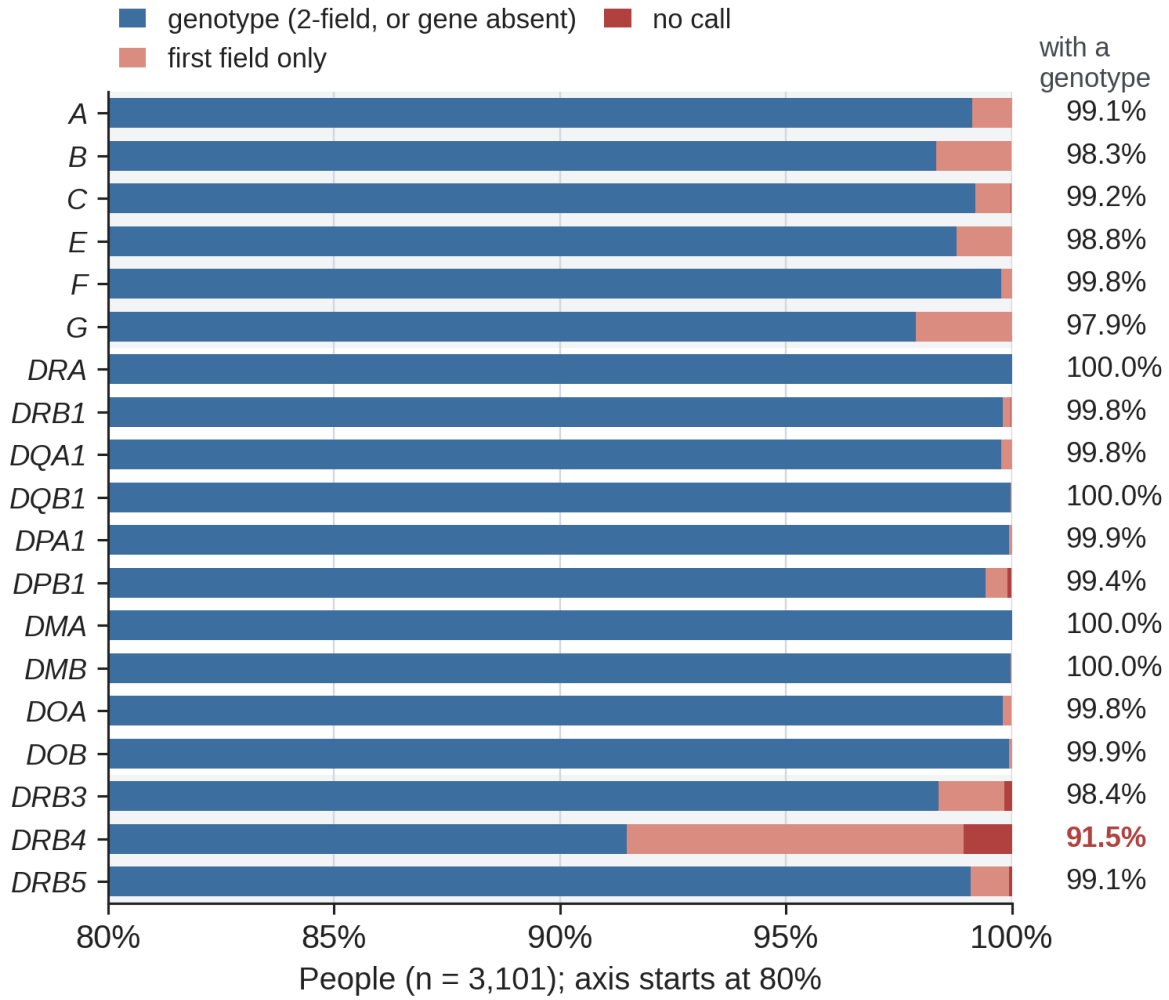
P group: alleles with identical protein over the peptide-binding exons, named after the lowest-numbered member with a P; an allele in no P group keeps its own name (Background). Null allele: an allele that makes no protein (suffix N). Copy: one of a person's two copies of a gene, one on each chromosome 6.

Before any comparison, both sides share population, genes, resolution, and frequency definition.

Abbreviations: 61KJPN, about 61,000 Japanese individuals; HLA, human leukocyte antigen; IPD-IMGT/HLA, Immuno Polymorphism Database – international ImMunoGeneTics information system, HLA section; ToMMo, Tohoku Medical Megabank Organization.

[1] Tadaka, S., Kawashima, J., Hishinuma, E., Saito, S., Okamura, Y., Otsuki, A., ... & Kinoshita, K. (2024). jMorp: Japanese Multi-Omics Reference Panel update report 2023. *Nucleic Acids Research, 52*(D1), D622–D632.
[2] Anthony Nolan Research Institute. (n.d.). *P groups*. HLA Nomenclature. Retrieved October 19, 2026, from https://hla.alleles.org/pages/p&g_groups/p_groups/

Completeness of Typing



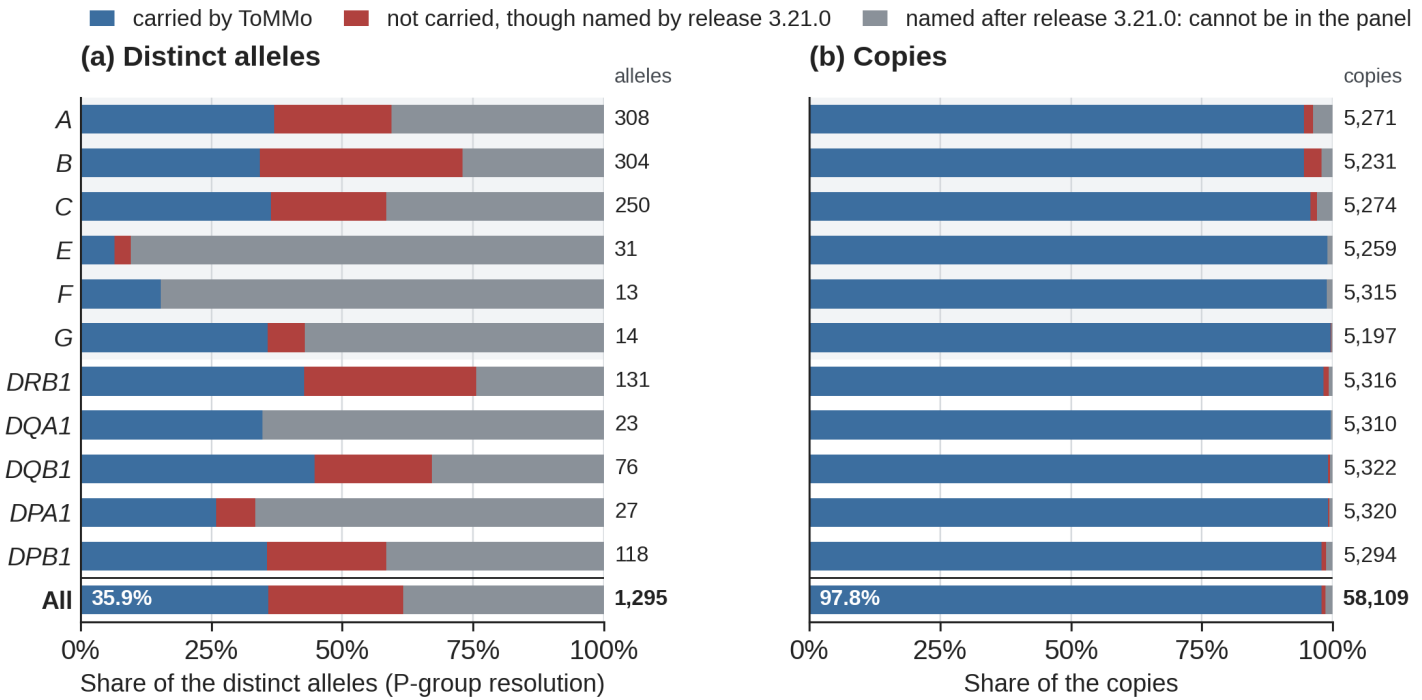
	n	of	%
Typing runs complete and validated	3,101	3,101	100.0%
Calls with a result (3,101 × 19 genes)	58,874	58,919	99.9%
Calls resolved to 2 fields	58,319	58,919	99.0%

Each bar is one gene, drawn from 80% to 100% of people. Blue: a genotype, i.e., a 2-field genotype, or *DRB3*, *DRB4*, or *DRB5* absent from both haplotypes, which is a result. *DRB4* is lowest (91.5%): its candidate alleles often differ only outside the reads' reach.

Every genome was typed, and 99.0% of the 58,919 calls reach 2-field resolution.

Abbreviations: HLA, human leukocyte antigen.

Presence of the Called Alleles in ToMMo 61KJPN



Not compared

DRA, DMA, DMB, DOA, DOB, and DRB5 have no entry in ToMMo.

DRB3 and DRB4 are excluded: ToMMo assigns a DRB3 allele to every haplotype although only a quarter carry the gene, and its DRB4 entries do not match its own DRB1. Their presence here follows DRB1, so they are also excluded from the association.

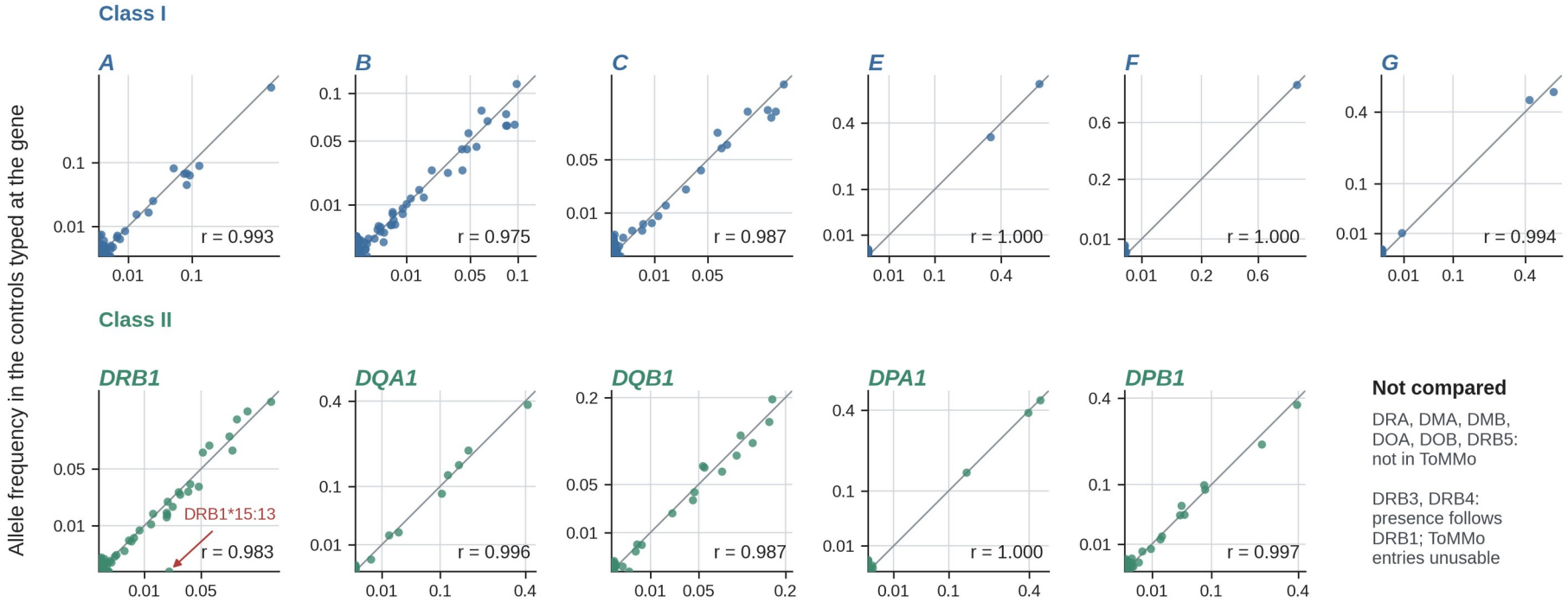
The 2,662 controls at the 11 genes ToMMo also reports, both sides harmonized as in the Method: alleles at P-group resolution, null alleles left out. (a) Each distinct allele counts once, however many controls carry it. (b) Each copy counts once, two per control typed at the gene. Blue: ToMMo carries the allele. Red: ToMMo does not, although the allele was named by release 3.21.0, as every ToMMo allele was. Gray: the allele was named after that release, so it cannot be in the panel whatever its true frequency. Alleles that only ToMMo carries are not counted here; “Concordance of Allele Frequencies with ToMMo 61KJPN” plots them.

Distinct alleles versus copies. Of the 1,295 distinct alleles called (at P-group resolution) across the 11 genes, 465 (35.9%) are also in ToMMo, yet they carry 97.8% of the 58,109 copies. The rest is a tail of rare alleles: those ToMMo could have contained carry 0.8% of the copies; those named after release 3.21.0 carry 1.4%.

97.8% of the copies called in the controls carry an allele that ToMMo also carries.

Abbreviations: 61KJPN, about 61,000 Japanese individuals; ToMMo, Tohoku Medical Megabank Organization.

Concordance of Allele Frequencies with ToMMo 61KJPN



One panel per gene. Each dot is one allele at P-group resolution: on the diagonal it is equally frequent in the 2,662 controls and in 61,424 Japanese individuals [1]. The dots are every allele found in either table (2,183 over the 11 genes, 465 of them in both); an allele one side lacks sits at 0 on that side, so a miscalled or missed allele lowers r. Both sides are compared at P-group resolution (Method), so the concordance checks the typing only to that resolution; the tested allele columns that share a P group are listed in “Notes: Tested Allele Columns That Share a P Group”. r is computed within each gene; all 11 compared genes have r ≥ 0.975.

One dot lies clearly off the diagonal: DRB1*15:13, marked in the *DRB1* panel. It is explained in “The One Dot off the Diagonal: DRB1*15:13”.

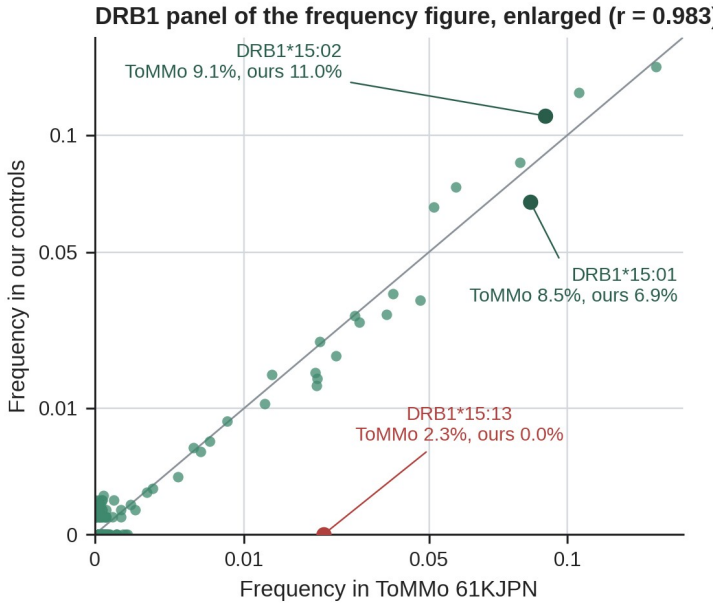
Allele frequencies in the controls are concordant with those of 61,424 Japanese individuals (r ≥ 0.975).

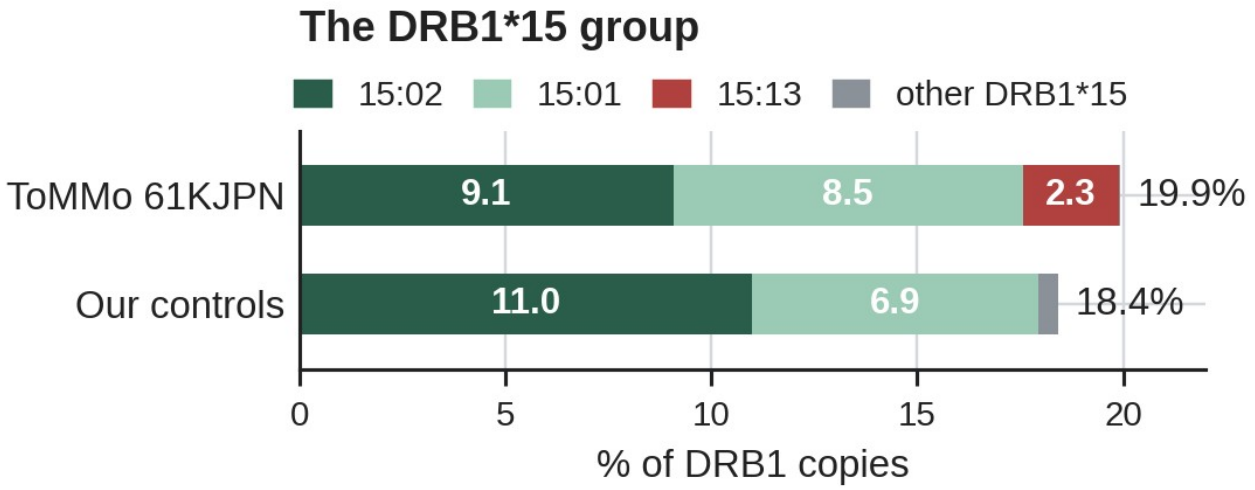
Abbreviations: 61KJPN, about 61,000 Japanese individuals; ToMMo, Tohoku Medical Megabank Organization.

[1] Tadaka, S., Kawashima, J., Hishinuma, E., Saito, S., Okamura, Y., Otsuki, A., ... & Kinoshita, K. (2024). jMorp: Japanese Multi-Omics Reference Panel update report 2023. *Nucleic Acids Research, 52*(D1), D622–D632.

The One Dot off the Diagonal: DRB1*15:13

What the dot is, and why	
What	ToMMo puts 2.3% of its *DRB1* copies on DRB1*15:13 (2,864), an allele named long before release 3.21.0, so ToMMo could carry it. Our controls: none.
Why	In IPD-IMGT/HLA 3.64.0, the dictionary HLA-HD used, it is DRB1*15:13Q (renamed in release 3.59.0), in no P group, identical in protein to DRB1*15:01 wherever both are known (219 positions), and absent from HLA-HD's frequency table: nothing lets HLA-HD prefer it, so these copies get another DRB1*15 name.
Where	Right: the DRB1*15 group totals are close (18.4% here, 19.9% in ToMMo); inside it, our DRB1*15:02 is 1.9 points higher and DRB1*15:01 1.5 points lower.
Verdict	A naming difference between dictionary releases, not copies lost by the typing.





A naming difference between dictionary releases, not copies lost by the typing.

Abbreviations: 61KJPN, about 61,000 Japanese individuals; HLA, human leukocyte antigen; IPD-IMGT/HLA, Immuno Polymorphism Database – international ImMunoGeneTics information system, HLA section; ToMMo, Tohoku Medical Megabank Organization.

[1] Tadaka, S., Kawashima, J., Hishinuma, E., Saito, S., Okamura, Y., Otsuki, A., ... & Kinoshita, K. (2024). jMorp: Japanese Multi-Omics Reference Panel update report 2023. *Nucleic Acids Research, 52*(D1), D622–D632.

Amino-Acid Residues: Rationale and Encoding

(a) Why residues as well as alleles

	5	6	7	8	**9**	10	11	12	13
A*02:01	M	R	Y	F	**F**	T	S	V	S
A*02:06	M	R	Y	F	**Y**	T	S	V	S
A*11:01	M	R	Y	F	**Y**	T	S	V	S
A*26:01	M	R	Y	F	**Y**	T	S	V	S
A*24:02	M	R	Y	F	**S**	T	S	V	S

9 Y one column

Three allele groups (A*02, A*11, A*26), one amino acid:
the residue column 9 Y tests these alleles together,
so a signal can be traced to one position.

A*02:01 and A*02:06 share the first field (A*02), not the amino acid.

(b) How a residue column is filled

One person: A*24:02 / A*11:01

amino acids from the IPD-IMGT/HLA protein alignment, not the reads

	5	6	7	8	**9**	10	11	12	13
A*24:02	M	R	Y	F	**S**	T	S	V	S
A*11:01	M	R	Y	F	**Y**	T	S	V	S

9 F	9 S	9 T	9 Y	
0	1	0	1	= 2

One column per amino acid seen at the position;
each cell counts the copies that carry it: 0, 1, or 2.

(c) When a cell is blank

One person in the cohort: A*24:02 / A*24:82

	180	181	182	183	**184**	185	186	187	188
A*24:02	Q	R	T	D	**P**	P	K	T	H
A*24:82	Q	R	T	*	*****	*	*	*	*

* = not sequenced (A*24:82 covers only the groove exons)

	184 A	184 P
True count the unsequenced copy is A or P	0 or 1	1 or 2
Known copy only counts the unsequenced copy as 0	0	1
Written: blank unknown, so left out of the test		

Real HLA-A sequences of the IPD-IMGT/HLA 3.64.0 protein alignment. (a) Position 9 lies on the floor of the peptide-binding groove; testing residues as well as alleles is the standard way to trace an HLA signal to single positions [1], and a residue column also counts the alleles too rare to be tested on their own. (b) Each 2-field allele is matched to its row of the alignment. (c) A person with no 2-field genotype at a gene is likewise blank in every column of that gene.

Residue columns pool alleles that share an amino acid; where the alignment cannot name one, the cell is blank.

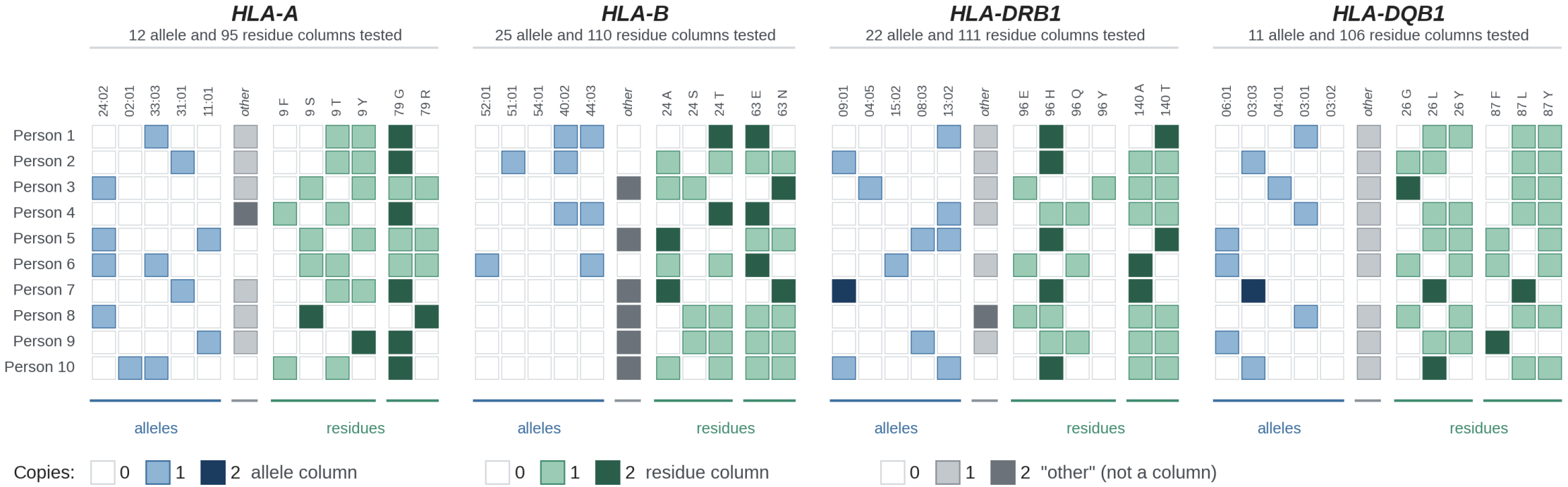
Abbreviations: HLA, human leukocyte antigen; IPD-IMGT/HLA, Immuno Polymorphism Database – international ImMunoGeneTics information system, HLA section.

[1] Sakaue, S., Gurajala, S., Curtis, M., Luo, Y., Choi, W., Ishigaki, K., ... & Raychaudhuri, S. (2023). Tutorial: A statistical genetics guide to identifying HLA alleles driving complex disease. *Nature Protocols, 18*(9), 2625–2641.

Encoding of the Markers for Association Testing

The matrix the association reads: full mainland cohort

10 of its 3,101 people × 4 of its 16 genes. Each column is one marker and each value the number of copies — the encoding the genome-wide scan uses for a SNP.



Real data. Column names: an allele column is the 2-field allele; a residue column is the position and the amino acid, so "9 F" is phenylalanine at position 9.

Shown per gene: the 5 most frequent of the allele columns tested, and two positions with every amino acid seen there as a column (a gap separates them).

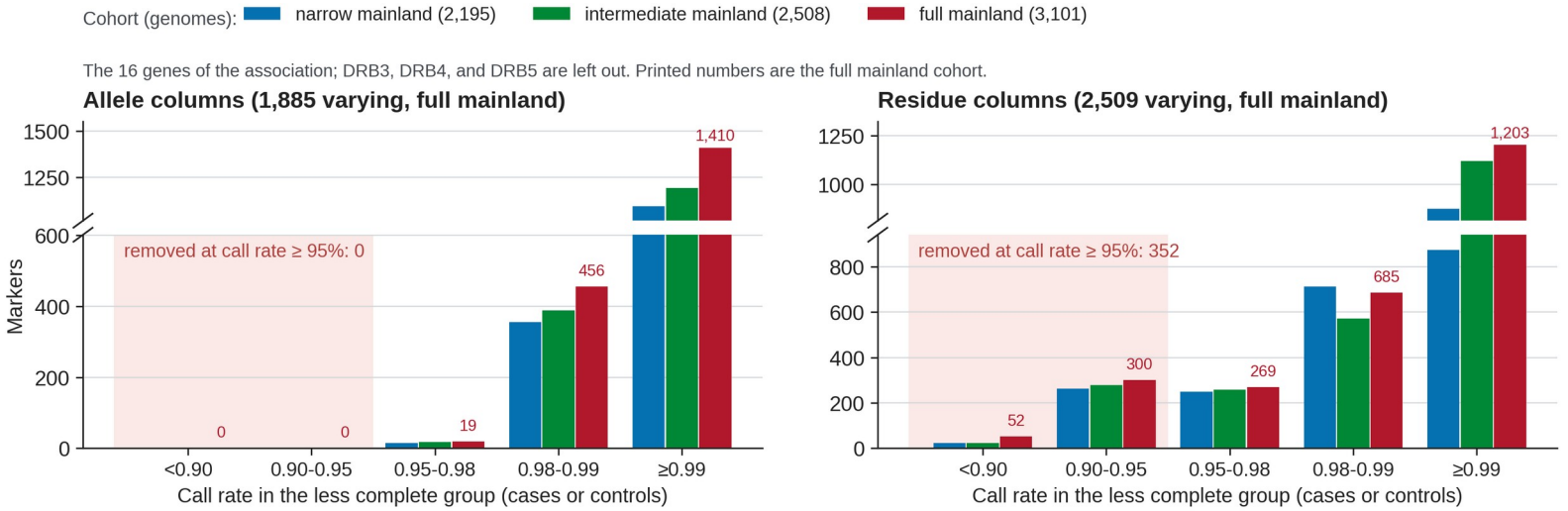
"other" is not a column of the matrix: it counts the person's remaining copies, of alleles not shown here or of alleles below the MAF floor, which have no column.

Per gene, each person therefore adds up to 2 copies: over the alleles shown plus "other", and over the residue columns at each position (one amino acid per copy).

Each column is one allele, or one amino acid at one position, coded as 0/1/2 copies like a SNP in the genome-wide scan.

Abbreviations: HLA, human leukocyte antigen; MAF, minor allele frequency.

Call Rate of the HLA Markers



Call-rate floor	Allele columns lost narrow / intermediate / full mainland	Residue columns lost narrow / intermediate / full mainland
≥ 95% (chosen)	0 / 0 / 0	285 / 301 / 352
≥ 98%	15 / 18 / 19	534 / 560 / 621
≥ 99%	371 / 407 / 475	1,247 / 1,132 / 1,306

Rationale for call rate ≥ 95%. (1) Comparable studies apply a call-rate floor to array SNPs before imputation (≥ 95% [1], ≥ 98% [2]); typed calls carry no imputation score, so call rate is the only completeness measure available. (2) 95% lies below every gene's completeness in either group (lowest *G* in the full mainland controls, 97.8%), so no gene is lost, and above the positions that IPD-IMGT/HLA leaves largely undetermined, which are removed.

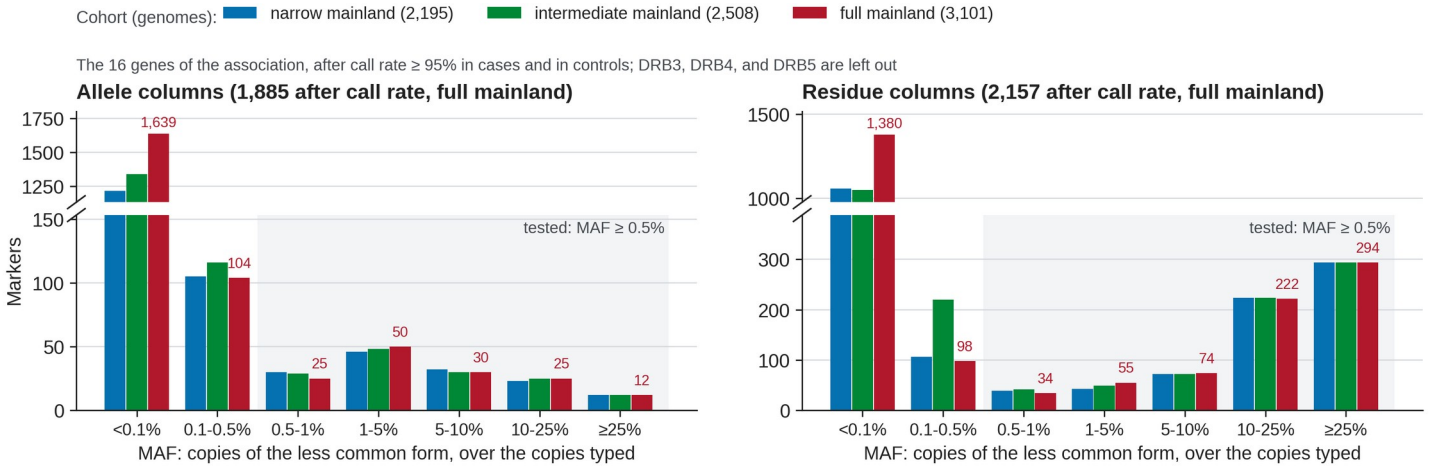
Call rate is measured in cases and in controls separately, and a marker must reach the floor in both: the sequencing platform and depth differ between the two groups. Losses are counted over the columns that still vary in each cohort (narrow / intermediate / full mainland): 1,465 / 1,599 / 1,885 allele columns and 2,121 / 2,251 / 2,509 residue columns.

Call rate ≥ 95% removes no allele column, only residue columns the alignment leaves undetermined.

Abbreviations: HLA, human leukocyte antigen; IPD-IMGT/HLA, Immuno Polymorphism Database – international ImMunoGeneTics information system, HLA section; MAF, minor allele frequency.

[1] Butler-Laporte, G., Farjoun, Y., Chen, Y., Hultström, M., Liang, K. Y. H., Nakanishi, T., ... & Richards, J. B. (2023). HLA allele-calling using multi-ancestry whole-exome sequencing from the UK Biobank identifies 129 novel associations in 11 autoimmune diseases. *Communications Biology, 6*(1), 1113.
[2] Sakaue, S., Yamaguchi, E., Inoue, Y., Takahashi, M., Hirata, J., Suzuki, K., ... & Okada, Y. (2021). Genetic determinants of risk in autoimmune pulmonary alveolar proteinosis. *Nature Communications, 12*(1), 1032.

Minor Allele Frequency of the HLA Markers



Cohort	Genomes	Cases	Controls	Copies at MAF 0.5% if every genome is typed	Expected in cases copies, with no association	Allele columns tested	Residue columns tested
narrow mainland	2,195	419	1,776	22	4.2	143	672
intermediate mainland	2,508	429	2,079	25	4.3	144	681
full mainland	3,101	439	2,662	31	4.4	142	679

Rationale for MAF ≥ 0.5%. (1) It is the floor of comparable HLA association studies [1]–[3] and lies within the 0.5–1% range recommended by [4]. (2) A marker at the floor is expected on 4.2–4.4 copies in cases (table above), close to the five per cell that a 2 × 2 table requires. (3) The distribution offers no natural cut point between 0.3% and 1%, so the floor is a choice about the fewest copies in cases that can be tested, not a feature of the data.

Applied to the markers left by the call-rate floor (narrow / intermediate / full mainland): 1,465 / 1,599 / 1,885 allele columns and 1,836 / 1,950 / 2,157 residue columns. **Copy**: one of a person's two gene copies (0, 1, or 2 per person). **MAF**: copies of the less common form over the copies typed.

Most HLA alleles are below 0.1%; 142 allele and 679 residue columns pass MAF ≥ 0.5% (full mainland).

Abbreviations: HLA, human leukocyte antigen; MAF, minor allele frequency.

[1] Luo, Y., Kanai, M., Choi, W., Li, X., Sakaue, S., Yamamoto, K., ... & Raychaudhuri, S. (2021). A high-resolution HLA reference panel capturing global population diversity enables multi-ancestry fine-mapping in HIV host response. *Nature Genetics, 53*(10), 1504–1516.
[2] Hirata, J., Hosomichi, K., Sakaue, S., Kanai, M., Nakaoka, H., Ishigaki, K., ... & Okada, Y. (2019). Genetic and phenotypic landscape of the major histocompatibilty complex region in the Japanese population. *Nature Genetics, 51*(3), 470–480.
[3] Sakaue, S., Yamaguchi, E., Inoue, Y., Takahashi, M., Hirata, J., Suzuki, K., ... & Okada, Y. (2021). Genetic determinants of risk in autoimmune pulmonary alveolar proteinosis. *Nature Communications, 12*(1), 1032.
[4] Sakaue, S., Gurajala, S., Curtis, M., Luo, Y., Choi, W., Ishigaki, K., ... & Raychaudhuri, S. (2023). Tutorial: A statistical genetics guide to identifying HLA alleles driving complex disease. *Nature Protocols, 18*(9), 2625–2641.

Marker Quality Control: Stepwise Record in Three Cohorts

Step	Rule	Genes narrow / intermediate / full mainland	Allele columns narrow / intermediate / full mainland	Residue columns narrow / intermediate / full mainland
Typed by HLA-HD	HLA-HD 1.7.1, IPD-IMGT/HLA 3.64.0; a residue the alignment leaves undetermined is blank	19 / 19 / 19	1,548 / 1,696 / 1,995	2,465 / 2,607 / 2,872
***DRB3*, *DRB4*, and *DRB5* excluded**	presence follows *DRB1*; excluded from the association	16 / 16 / 16	1,465 / 1,599 / 1,885	2,121 / 2,251 / 2,509
Call rate ≥ 95%	≥ 95% of cases and ≥ 95% of controls have a genotype at the marker	16 / 16 / 16	1,465 / 1,599 / 1,885	1,836 / 1,950 / 2,157
MAF ≥ 0.5%	the frequency floor	16 / 16 / 16	143 / 144 / 142	672 / 681 / 679

A column is counted while it still varies in that cohort — at least one copy of its less common form. Every genome of every cohort is typed (narrow / intermediate / full mainland: 2,195 / 2,508 / 3,101), so no sample is lost at any step; the rules act on the columns only. Both matrices are files written by the pipeline for each cohort (results/06.assoc_inputs), and the association reads them as they stand.

Matrix exported by the typing (19 genes)
3,101 people × **1,995** allele columns
3,101 people × **2,872** residue columns



the rules above

Matrix the association reads (full mainland cohort)
3,101 people × **142** allele columns
3,101 people × **679** residue columns

No sample is lost: every rule acts on columns only.

Abbreviations: HLA, human leukocyte antigen; IPD-IMGT/HLA, Immuno Polymorphism Database – international ImMunoGeneTics information system, HLA section; MAF, minor allele frequency.

Allele Columns Retained at Each Step, by Gene

Gene	Typed by HLA-HD	*DRB3*, *DRB4*, and *DRB5* excluded	Call rate ≥ 95% in cases and in controls	MAF ≥ 0.5%	Copies covered, % tested-allele copies / all copies typed
A	311 / 340 / 411	311 / 340 / 411	311 / 340 / 411	11 / 11 / 12	86.6 / 86.3 / 86.3
B	319 / 351 / 411	319 / 351 / 411	319 / 351 / 411	25 / 25 / 25	85.4 / 85.0 / 84.6
C	281 / 303 / 355	281 / 303 / 355	281 / 303 / 355	14 / 15 / 15	88.3 / 88.9 / 88.5
E	37 / 38 / 45	37 / 38 / 45	37 / 38 / 45	3 / 3 / 3	98.8 / 98.8 / 98.7
F	18 / 19 / 20	18 / 19 / 20	18 / 19 / 20	4 / 4 / 2	99.2 / 99.2 / 98.2
G	15 / 18 / 19	15 / 18 / 19	15 / 18 / 19	3 / 3 / 3	98.9 / 98.8 / 98.8
DRA	5 / 6 / 6	5 / 6 / 6	5 / 6 / 6	4 / 4 / 4	>99.9 / >99.9 / >99.9
DRB1	116 / 125 / 156	116 / 125 / 156	116 / 125 / 156	22 / 22 / 22	95.8 / 95.6 / 95.2
DRB3	23 / 27 / 32	— / — / —	— / — / —	— / — / —	—
DRB4	46 / 54 / 60	— / — / —	— / — / —	— / — / —	—
DRB5	14 / 16 / 18	— / — / —	— / — / —	— / — / —	—
DQA1	44 / 45 / 49	44 / 45 / 49	44 / 45 / 49	14 / 14 / 14	98.1 / 98.0 / 98.1
DQB1	83 / 93 / 117	83 / 93 / 117	83 / 93 / 117	12 / 12 / 11	97.3 / 97.2 / 96.6
DPA1	68 / 73 / 85	68 / 73 / 85	68 / 73 / 85	8 / 8 / 8	97.5 / 97.4 / 97.4
DPB1	141 / 158 / 181	141 / 158 / 181	141 / 158 / 181	12 / 12 / 12	94.3 / 94.1 / 94.1
DMA	8 / 8 / 8	8 / 8 / 8	8 / 8 / 8	3 / 3 / 3	99.8 / 99.9 / 99.9
DMB	7 / 8 / 8	7 / 8 / 8	7 / 8 / 8	4 / 4 / 4	99.6 / 99.6 / 99.6
DOA	8 / 9 / 9	8 / 9 / 9	8 / 9 / 9	1 / 1 / 1	99.4 / 99.4 / 99.3
DOB	4 / 5 / 5	4 / 5 / 5	4 / 5 / 5	3 / 3 / 3	>99.9 / 99.9 / 99.9
Total	**1,548 / 1,696 / 1,995**	**1,465 / 1,599 / 1,885**	**1,465 / 1,599 / 1,885**	**143 / 144 / 142**	**96.2 / 96.1 / 96.0**

Each cell: narrow / intermediate / full mainland. A gene's alleles share its call rate in each group, so none is removed; uncovered copies count in residue columns.

MAF ≥ 0.5% keeps 142 of 1,885 allele columns, which carry 96.0% of the copies (full mainland).

Abbreviations: HLA, human leukocyte antigen; MAF, minor allele frequency.

Residue Columns Retained at Each Step, by Gene

Gene	Typed by HLA-HD	*DRB3*, *DRB4*, and *DRB5* excluded	Call rate ≥ 95% in cases and in controls	MAF ≥ 0.5%
A	455 / 480 / 531	455 / 480 / 531	327 / 349 / 383	86 / 95 / 95
B	302 / 327 / 375	302 / 327 / 375	246 / 265 / 290	110 / 110 / 110
C	314 / 342 / 401	314 / 342 / 401	221 / 242 / 290	68 / 68 / 68
E	67 / 67 / 79	67 / 67 / 79	67 / 67 / 79	4 / 4 / 4
F	34 / 36 / 37	34 / 36 / 37	34 / 36 / 37	6 / 6 / 4
G	138 / 142 / 144	138 / 142 / 144	138 / 142 / 144	6 / 6 / 6
DRA	8 / 10 / 10	8 / 10 / 10	6 / 8 / 8	4 / 4 / 4
DRB1	192 / 195 / 212	192 / 195 / 212	192 / 195 / 212	111 / 111 / 111
DRB3	73 / 75 / 75	— / — / —	— / — / —	— / — / —
DRB4	177 / 185 / 191	— / — / —	— / — / —	— / — / —
DRB5	94 / 96 / 97	— / — / —	— / — / —	— / — / —
DQA1	125 / 126 / 134	125 / 126 / 134	123 / 124 / 132	91 / 91 / 91
DQB1	177 / 193 / 216	177 / 193 / 216	177 / 193 / 216	106 / 106 / 106
DPA1	104 / 109 / 128	104 / 109 / 128	104 / 109 / 128	18 / 18 / 18
DPB1	155 / 168 / 186	155 / 168 / 186	151 / 164 / 182	42 / 42 / 42
DMA	15 / 15 / 15	15 / 15 / 15	15 / 15 / 15	6 / 6 / 6
DMB	15 / 17 / 17	15 / 17 / 17	15 / 17 / 17	10 / 10 / 10
DOA	14 / 16 / 16	14 / 16 / 16	14 / 16 / 16	0 / 0 / 0
DOB	6 / 8 / 8	6 / 8 / 8	6 / 8 / 8	4 / 4 / 4
Total	**2,465 / 2,607 / 2,872**	**2,121 / 2,251 / 2,509**	**1,836 / 1,950 / 2,157**	**672 / 681 / 679**

Each cell: narrow / intermediate / full mainland; a column is counted while it still varies in that cohort. Residue columns survive the frequency floor more often than allele columns, because many alleles share an amino acid. At the positions kept, the residue columns cover 99.97% of the copies.

Full mainland: call rate removes 352 residue columns, 344 in *A*, *B*, *C*; MAF ≥ 0.5% keeps 679 of 2,157.

Abbreviations: HLA, human leukocyte antigen; IPD-IMGT/HLA, Immuno Polymorphism Database – international ImMunoGeneTics information system, HLA section; MAF, minor allele frequency.

Summary and Conclusions

3,101	99.0%	r ≥ 0.975	142	679
whole genomes typed at 19 HLA genes; all 3,101 runs validated	of 58,919 calls resolved to 2 fields (99.9% with a result)	with ToMMo 61KJPN across 11 genes, in the 2,662 controls	allele columns for the association, full mainland cohort	residue columns for the association, full mainland cohort

	Typing quality	Markers for association testing
Scope	3,101 genomes (439 cases, 2,662 controls), 19 genes	16 genes; *DRB3*, *DRB4*, and *DRB5* excluded
Criterion	completeness; presence and frequency of the controls' alleles in ToMMo 61KJPN (P-group resolution)	call rate ≥ 95% in cases and in controls, then MAF ≥ 0.5%
Outcome	every genome typed; 97.8% of control copies carry an allele ToMMo carries; r = 0.975–1.000 across 11 genes	143 / 144 / 142 allele and 672 / 681 / 679 residue columns in the narrow / intermediate / full mainland cohorts
Caveat	the concordance checks the typing to P-group resolution only	31 of the 142 allele columns tested in the full mainland cohort share a P group with another tested column; ToMMo sees only their sum (“Notes: Tested Allele Columns That Share a P Group”)

Methods statement. HLA alleles were typed from whole-genome sequencing reads with HLA-HD 1.7.1 against IPD-IMGT/HLA 3.64.0. Allele frequencies in the 2,662 AGP3K controls, compared at P-group resolution with null alleles excluded, were highly concordant with those of 61,424 Japanese individuals in the ToMMo 61KJPN reference panel (Pearson r = 0.975–1.000 across 11 genes). 2-field alleles and amino-acid residues of 16 genes (*DRB3*, *DRB4*, and *DRB5* excluded) were encoded as 0/1/2 dosages, a residue left undetermined by the reference alignment being set to missing, and retained for testing at a call rate of ≥ 95% in both cases and controls and a minor allele frequency of ≥ 0.5%.

The HLA genotypes are complete, concordant with the Japanese population, and suitable for association testing.

Abbreviations: 61KJPN, about 61,000 Japanese individuals; HLA, human leukocyte antigen; ToMMo, Tohoku Medical Megabank Organization.

Notes: How Our Alleles Map onto the ToMMo Table

Both tables are read with one P-group file, IPD-IMGT/HLA 3.64.0: an allele in a P group is replaced by its group, an allele in no P group keeps its own name, and null alleles are left out (Method). ToMMo publishes its table in this form: a name ending in P is a P group, a name without P an allele in no P group. Our 2-field calls change only where two or more of them fall in the same P group: they are merged into it.

Example, *DQA1*. Our controls carry DQA1*03:01 (8.5%), DQA1*03:03 (14.9%), DQA1*03:02 (14.8%) and 4 rarer alleles of the same P group (0.1%). Merged, they are DQA1*03:01P on 38.4% of the copies; ToMMo lists only DQA1*03:01P, on 40.9%.

Gene	**ToMMo entries** P groups / alleles in no P group	**Our 2-field alleles** in the 2,662 controls	**Merged** alleles sharing a P group → P groups	**Our alleles at P-group resolution** compared with ToMMo	**In both tables** blue bars in "Presence of the Called Alleles in ToMMo 61KJPN"	**In either table** dots in "Concordance of Allele Frequencies with ToMMo 61KJPN"
A	74 / 273	376	76 → 8	308	114	541
B	125 / 205	374	92 → 22	304	104	530
C	76 / 193	311	73 → 12	250	91	428
E	2 / 0	39	10 → 2	31	2	31
F	2 / 0	17	5 → 1	13	2	13
G	3 / 4	17	5 → 2	14	5	16
DRB1	72 / 131	146	26 → 11	131	56	278
DQA1	8 / 1	45	27 → 5	23	8	24
DQB1	34 / 52	111	47 → 12	76	34	128
DPA1	7 / 4	77	55 → 5	27	7	31
DPB1	37 / 50	169	68 → 17	118	42	163
Total	**440 / 913**	**1,682**	**484 → 97**	**1,295**	**465**	**2,183**

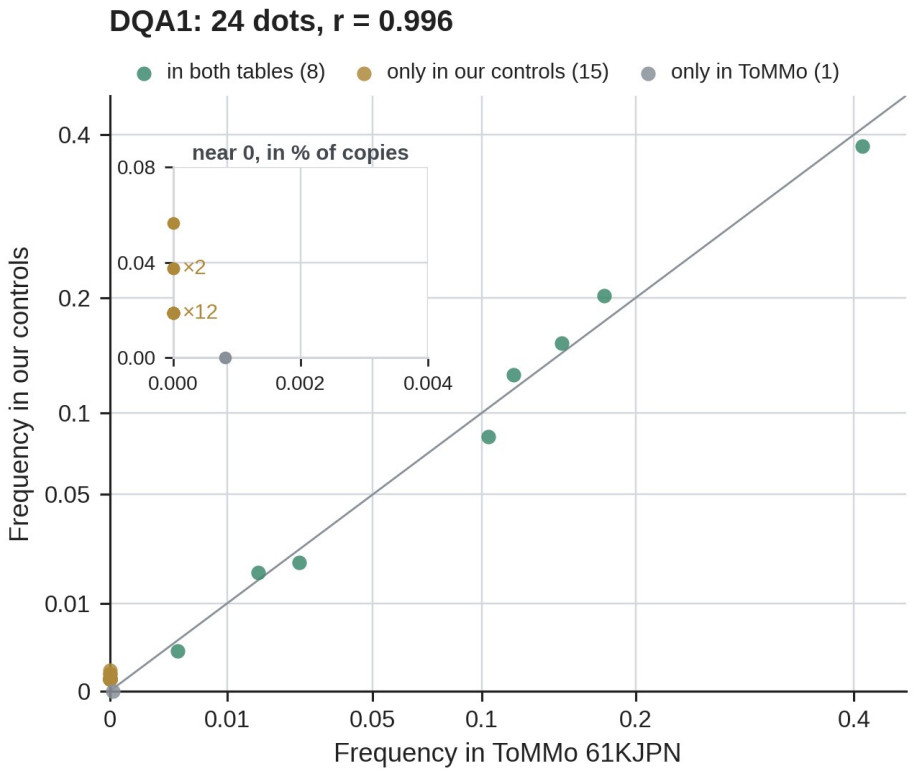
At P-group resolution, 484 of our 1,682 2-field alleles merge into 97 P groups, leaving 1,295.

Abbreviations: HLA, human leukocyte antigen; IPD-IMGT/HLA, Immuno Polymorphism Database – international ImMunoGeneTics information system, HLA section; ToMMo, Tohoku Medical Megabank Organization.

Notes: One Gene through the Comparison, DQA1

Every *DQA1* allele at P-group resolution found in either table, with its share of the copies in the 2,662 controls and in ToMMo (null alleles left out). The figure draws them as “Concordance of Allele Frequencies with ToMMo 61KJPN” does.

Allele P-group resolution	**Our 2-field alleles merged into it** DQA1*…, copies in the controls	**Controls** % of copies	**ToMMo** % of copies
DQA1*03:01P	03:03 (792), 03:02 (788), 03:01 (452), 03:18 (3), 03:54 (1), 03:81 (1), 03:85 (1)	38.4%	40.9%
DQA1*01:03P	01:03 (1,066), 01:176 (4), 01:102 (1)	20.2%	17.6%
DQA1*01:02P	01:02 (821), 01:100 (4), 01:104 (2), 01:63 (2), 01:191 (1)	15.6%	14.8%
DQA1*01:01P	01:01 (362), 01:04 (291), 01:05 (25), 01:157 (5), 01:169 (2)	12.9%	11.8%
DQA1*05:01P	05:05 (167), 05:03 (84), 05:07 (79), 05:08 (55), 05:01 (38), 05:06 (13), 05:48 (9)	8.4%	10.4%
DQA1*04:01P	04:01 (114)	2.1%	2.6%
DQA1*06:01P	06:01 (97)	1.8%	1.6%
DQA1*02:01P	02:01 (11)	0.2%	0.3%
15 only in our controls	01:77 (3), 03:49 (2), 03:94Q (2), and 12 more with 1 copy each	0.02%–0.06% each	0
DQA1*01:06 only in ToMMo	none	0	1 copy



***DQA1* through the comparison.** Our controls carry 45 2-field alleles; 27 of them merge into 5 P groups, leaving 23 at P-group resolution. ToMMo lists 9. 8 are in both tables, the blue part of “Presence of the Called Alleles in ToMMo 61KJPN”; 24 are in either table, the 24 dots of “Concordance of Allele Frequencies with ToMMo 61KJPN”, over which r = 0.996.

Only 8 *DQA1* alleles are in both tables, but r is computed over all 24 found in either.

Abbreviations: 61KJPN, about 61,000 Japanese individuals; HLA, human leukocyte antigen; ToMMo, Tohoku Medical Megabank Organization.

Notes: Tested Allele Columns That Share a P Group

In 16 P groups the full mainland matrix tests more than one allele column (44 of its 142). Their alleles share the protein over the peptide-binding exons and differ only outside them; ToMMo lists each such group as one entry, so it can confirm only their sum. Per cohort (narrow / intermediate / full mainland): 16 / 16 / 16 groups and 45 / 45 / 44 columns; in the genes ToMMo reports, 12 / 12 / 12 groups and 32 / 32 / 31 columns.

Gene	ToMMo entry % of ToMMo's copies of the gene	Our alleles in this P group % of the 2,662 controls' copies of the gene: tested columns, then the others	Our sum % of copies
B	B*07:02P (4.8%)	B*07:02 (5.0%), B*07:381 (0.7%); 5 others (0.1%)	5.7%
B	B*40:01P (5.5%)	B*40:01 (3.7%), B*40:379 (0.6%); 4 others (0.1%)	4.5%
B	B*44:03P (6.0%)	B*44:03 (6.5%), B*44:302 (1.1%); 5 others (0.3%)	8.0%
C	C*08:01P (7.2%)	C*08:01 (5.8%), C*08:22 (0.6%); 2 others (0.3%)	6.7%
E	E*01:01P (33.9%)	E*01:01 (30.5%), E*01:12 (1.4%); 4 others (0.1%)	32.0%
F	F*01:01P (>99.9%)	F*01:01 (97.2%), F*01:05 (1.1%); 3 others (0.5%)	98.8%
DRA	gene not reported	DRA*01:01 (58.9%), DRA*01:02 (31.3%), DRA*01:03 (8.2%), DRA*01:05 (1.4%); 2 others (<0.1%)	100.0%
DQA1	DQA1*01:01P (11.8%)	DQA1*01:01 (6.8%), DQA1*01:04 (5.5%); 3 others (0.6%)	12.9%
DQA1	DQA1*03:01P (40.9%)	DQA1*03:01 (8.5%), DQA1*03:03 (14.9%), DQA1*03:02 (14.8%); 4 others (0.1%)	38.4%
DQA1	DQA1*05:01P (10.4%)	DQA1*05:01 (0.7%), DQA1*05:05 (3.1%), DQA1*05:03 (1.6%), DQA1*05:07 (1.5%), DQA1*05:08 (1.0%); 2 others (0.4%)	8.4%
DPA1	DPA1*01:03P (39.4%)	DPA1*01:03 (36.4%), DPA1*01:33 (1.0%), DPA1*01:95 (0.7%); 23 others (0.8%)	38.8%
DPA1	DPA1*02:02P (45.6%)	DPA1*02:02 (42.5%), DPA1*02:33 (0.9%), DPA1*02:85 (0.6%), DPA1*02:07 (0.7%); 11 others (0.5%)	45.1%
DPB1	DPB1*05:01P (39.4%)	DPB1*05:01 (35.3%), DPB1*135:01 (1.3%); 6 others (0.4%)	37.0%
DMA	gene not reported	DMA*01:01 (83.2%), DMA*01:02 (15.5%), DMA*01:03 (1.1%); 4 others (0.1%)	>99.9%
DMB	gene not reported	DMB*01:01 (49.4%), DMB*01:03 (28.2%), DMB*01:02 (20.9%); 2 others (0.3%)	98.9%
DOB	gene not reported	DOB*01:01 (78.0%), DOB*01:02 (16.5%), DOB*01:04 (5.4%); 1 other (0.1%)	>99.9%

ToMMo sees each of these P groups as one entry; the association matrix tests its alleles apart.

Abbreviations: HLA, human leukocyte antigen; ToMMo, Tohoku Medical Megabank Organization.